Chapter - 4

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che-50

TU, 2017, 2018(t), 2022

(*) which complex of the following pairs has a large Δ value? Explain your answer?

(i) $[Cr(NH_3)_6]^{3-}$ and $[CrCl_6]^{3-}$

(ii) $[Co(H_2O)_6]^{3+}$ and $[Rh(H_2O)_6]^{3+}$

(iii) $[Cr(H_2O)_6]^{3+}$ and $[Cr(H_2O)_6]^{2+}$

⇒ NH_3 is stronger ligand than Cl^- & stronger ligand gives large Δ value. so $[Cr(NH_3)_6]^{3-}$ has large Δ value.

(ii) Co and Rh are the element of same group. Atomic number of Co(27) is less than Rh(45). Δ value increase with the increase of atomic number in the same group. so $[Rh(H_2O)_6]^{3+}$ has larger Δ value.

(iii) Δ value increases with increasing of oxidation state of metal ion. So $[Cr(H_2O)_6]^{3+}$ has larger Δ value.

(*) TU What are high spin & low spin complexes?

⇒ high spin: are the types of coordination complexes which have more unpaired e^- & are associated with weak field ligands. It occurs when Δ<P. Ex. complexes with ligands

like H_2O, F^-, Cl^- etc.

low spin complexes: are types of coordination complexes which have fewer unpaired e^- & are associated with ~~weak~~ strong field ligand. It occurs when $\Delta > P$. Ex. complexes with ligands like CN^-, CO (strong field ligand).

2018, 2019(t)
Which complex have larger Δ value?

(I) $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$

(II) $[Co(NH_3)_6]^{3+}$ and $[Rh(NH_3)_6]^{3+}$

⇒ (I) In both complex Fe has oxidation number +2. But as CN^- is stong field ligand than H_2O ligand. so $[Fe(CN)_6]^{4-}$ has larger Δ value.

(II) Co(27) has less atomic number than Rh(45) though they posses same group number. so $[Rh(NH_3)_6]^{3+}$ has larger Δ value.

2017
Calculate CFSE for a d^8 ion such as Ni^{2+} in octahedral & tetrahedral complexes. use units of Δ. in both cases. which is most stable?

⇒ Ni atom posses 3d orbital & d^8 ion means d orbital

contain 8 electron.

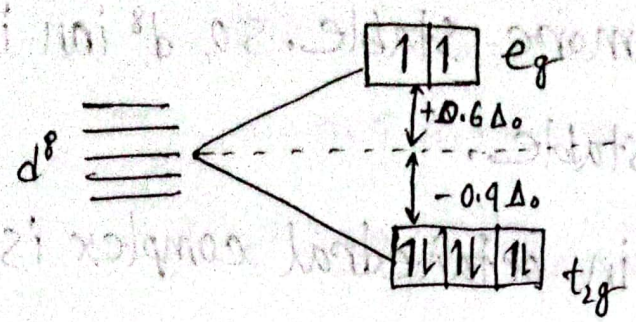


Fig: d^8 ion in octahedral field

electronic configuration $\Rightarrow d^8 \rightarrow t_{2g}^6 e_g^2$

$\therefore$ CFSE $= (-0.4 \times 6 + 0.6 \times 2)\Delta_o = -1.2\Delta_o$

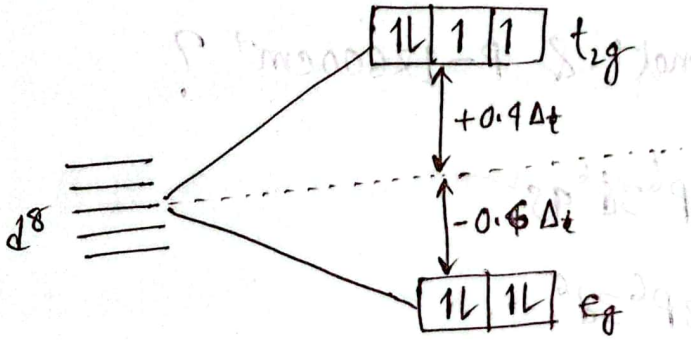


fig: d^8 ion in tetrahedral field

$\therefore$ electronic configuration $\Rightarrow d^8 \rightarrow e_g^4 t_{2g}^4$

$\therefore$ CFSE $= (-0.6 \times 4 + 0.4 \times 4)\Delta_t = -0.8\Delta_t$

$= -0.8 \times \frac{4}{9}\Delta_o \quad [\because \Delta_t = \frac{4}{9}\Delta_o]$

$= -0.35\Delta_o$

$\Rightarrow$ In octahedral field 6 ligands are attached to the d^8 ion & in tetrahedral field 4 ligands are attached to the d^8 ions.

the more ligands are attached to the central metal atom, the complex becomes more stable. so, d^8 ion in octahedral field is more stable.

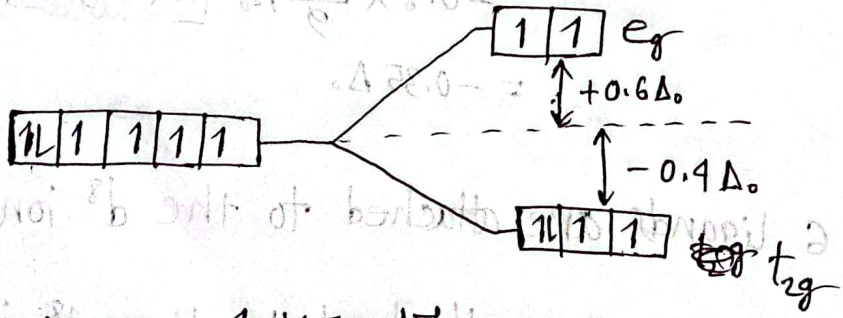
Also CFSE value of d^8 ions in octahedral complex is greater than that of tetrahedral complex. Greater the CFSE value more it will be stable. so, d^8 ion in octahedral complex is more stable.

(8) 2018, 2020, 2016 calculate CFSE for high spin $[Fe(H_2O)_6]^{2+}$ complex in KJmol^{-1}, Given, $\Delta_o = 10400$ KJmol^{-1} & $P = 17600\,cm^{-1}$?

$\Rightarrow$ $Fe(26) \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$

$Fe^{2+} \longrightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$

so, in $[Fe(H_2O)_6]^{2+}$ compound Fe^{2+} has d^6 configuration. since H_2O is weak field ligand, it forms high spin complex. And it has six ligand so it is octahedral in ~~nature~~ structure



we know, 1 KJmol^{-1} = 83.7 cm^{-1}

$\therefore CFSE = [(-0.4 \times 4) + (0.6 \times 2)]\Delta_o + P$

$= -0.4\,\Delta_o + P = -0.4 \times 10400 + \frac{17600}{83.7}$ [$\because 1 KJmol^{-1} = 83.7 cm^{-1}$]

$= -3949.73\ KJmol^{-1}$ Ans

2020, 2018, 2016

What is pairing energy?

⇒ Mean pairing energy (P) is the energy which is required to pair two e^- against electron-electron repulsion in the same orbital. P is generally expressed in cm^{-1}. pairing energy depends on the principal energy level (n) of d-electrons. Total pairing energy for m electron pairs = $mP\ cm^{-1}$.

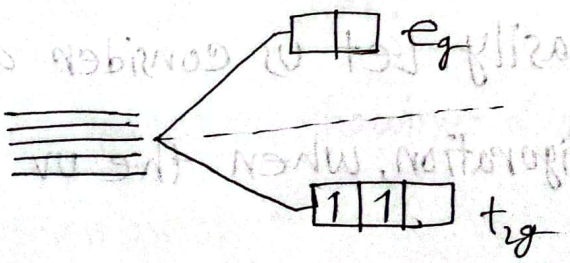
2020, 2016

Calculate CFSE for $[V(H_2O)_6]^{3+}$ in $KJmol^{-1}$. Given $\Delta_o = 18900\ KJmol^{-1}$?

⇒ $V(23) \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$

$V^{3+} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^0$

So, in $[V(H_2O)_6]^{3+}$ compound V^{3+} has d^2 configuration which is same for both high & low spin.



$\therefore CFSE = [(-0.4 \times 2) + (0.6 \times 0)]\Delta_o$

$= -0.8\Delta_o = -0.8 \times 18900\ KJmol^{-1} = -15120\ KJmol^{-1}$

⊛ 2018, 2020, 2016, 2021, 2018(t), 2017
What is crystal field stabilization energy?

⇒ CFSE refers to the energy difference btwn the energy of the e^- configuration of the ligand & the energy of e^- configuration of isotopic field. It depends on the amount of the central metal's e^- presen in the t_{2g} and e_g orbitals. denoted by Δ.

$CFSE = E_{ligand\ field} - E_{isotropic\ field}$

⊛ 2016, 2020
Define crystal field splitting? How would you measure it experimentally?

⇒ CFS refers to the phenomenon where the degenerate d-orbitals of a transition metal ion experience a change in energy levels when surrounded by ligands in a coordination complex.

⇒ By the electronic spectroscopy (UV), we can find the valve of CFS energy easily. Let us consider a complex $[Ti(H_2O)_6]^{3+}$ which is d^1 configuration, when the uv light passe

through this $[Ti(H_2O)_6]^{3+}$ we found single broad peak with a maximum at 20300 cm^{-1} due to the d-d transition.

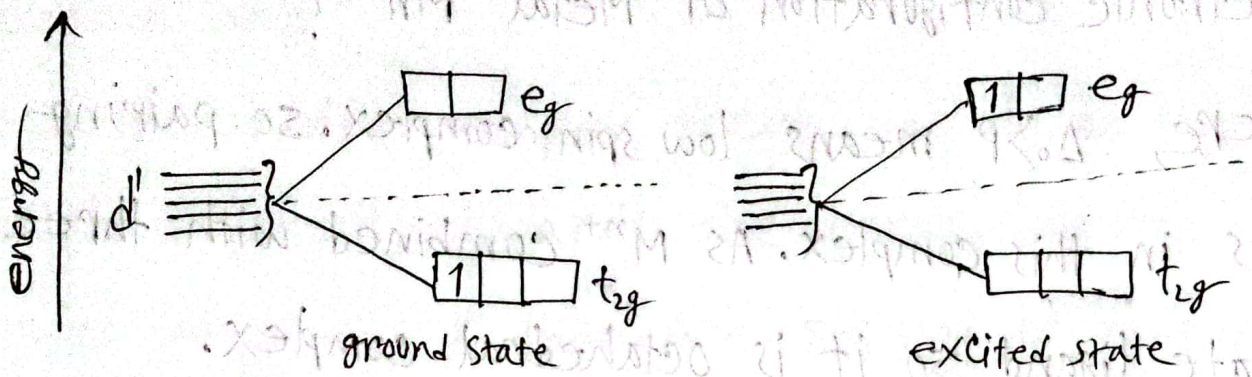
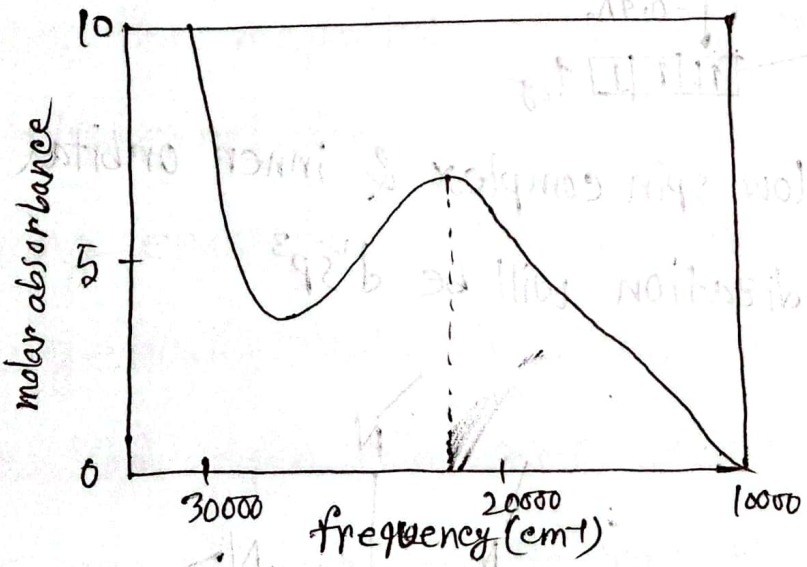


Fig: $[Ti(H_2O)_6]^{3+}$ ions ground state & excited state

The electronic spectrum for $[Ti(H_2O)_6]^{3+}$ is,



above method is the most convenient way of measuring Δ_o values.

2020
A metal M^{n+} having d^1 with 3 bidentate ligands to form complex. Assuming $\Delta_o > P$.

① Draw the diagram showing d-orbital splitting during this complex formation?

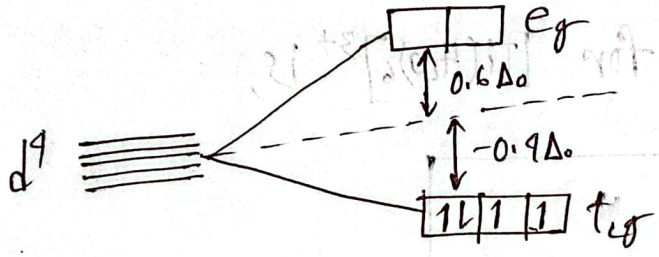
CamScanner

(ii) what type of hybridization will M^{n+} have?

(iii) Name the type of isomerism exibited by this complex

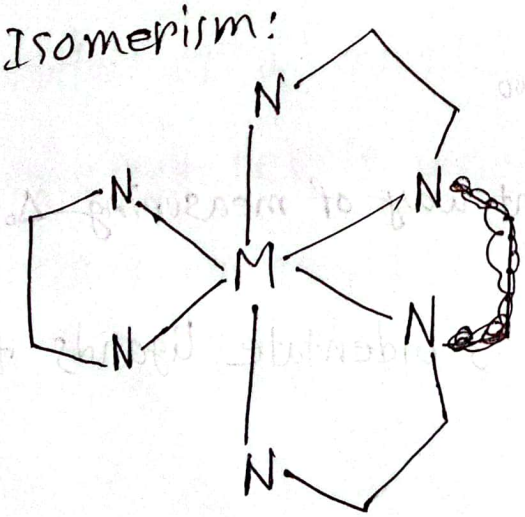
(iv) electronic configuration of Metal Mn^+ ?

⇒ (i) Here, $\Delta_o > P$ means low spin complex. so pairing occurs in this complex. As M^{n+} combined with three bidentate ligand so it is octahedral complex.

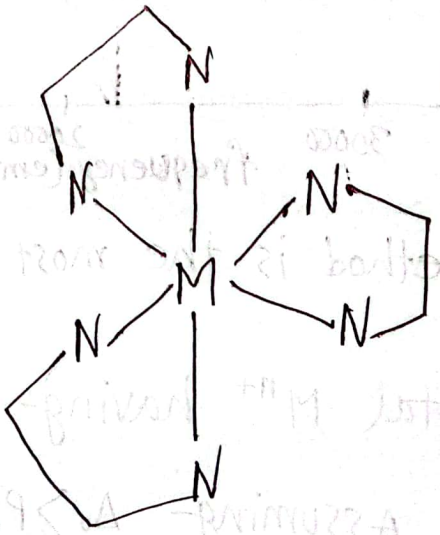


(ii) AS $\Delta_o > P$, this is low spin complex & inner orbital complex. so its hybridization will be d^2sp^3

(iii) Isomerism:



d-form



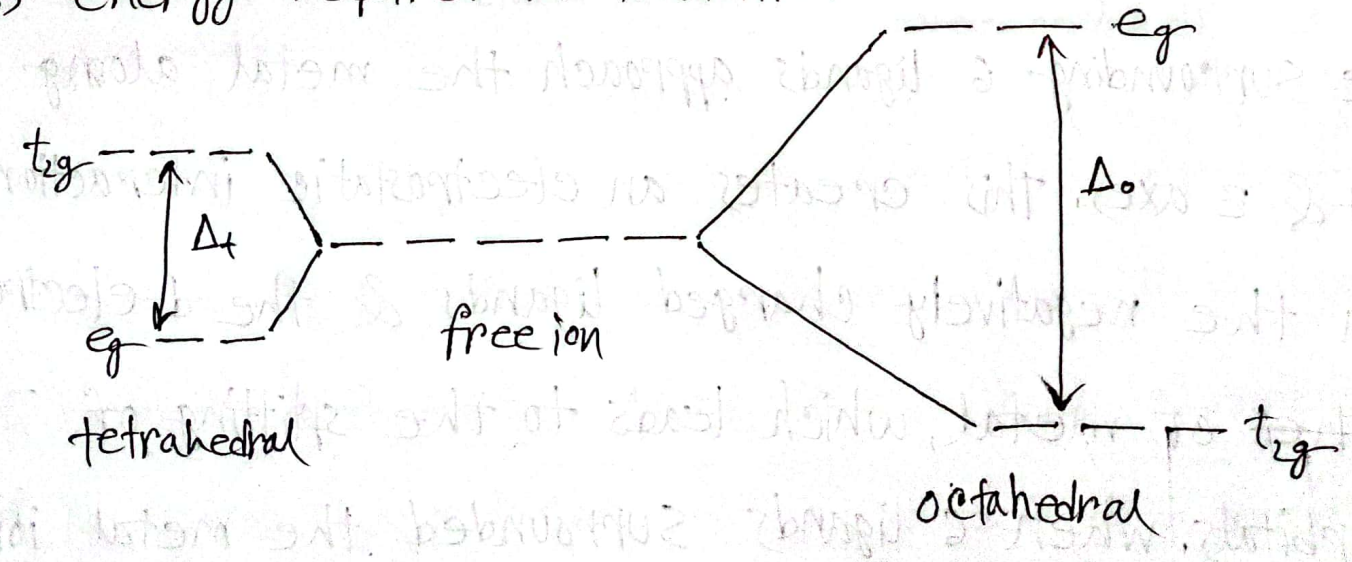
l-form

(iv) $Mn^{+}(24) \longrightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

2020, 2017

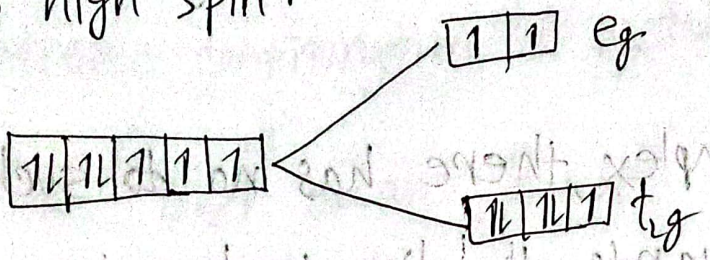
(x) Tetrahedral complexes are usually high spin complex- explain why?

⇒ In tetrahedral complex, there has no such d-orbital which is directly towards the ligands, lowering the $e^- - e^-$ repulsion & therefore splitting of energy is less compare to the octahedral field & less number of ligands produces less crystal field splitting energy. since the magnitude of CFSE (crystal field splitting energy) in tetrahedral field is small & always less than the pairing energy. so the pairing of e^- will never be energetically favourable. Thus all the tetrahedral complexes are high spin complexes, because of d-d transition, as less energy required for transition. & $\Delta_t = \frac{4}{9}\Delta_o$.



⊛ 2024, 2017, 2018(t) Find out the CFSE energy for d^7, both as high spin & low-spin in octahedral complex?

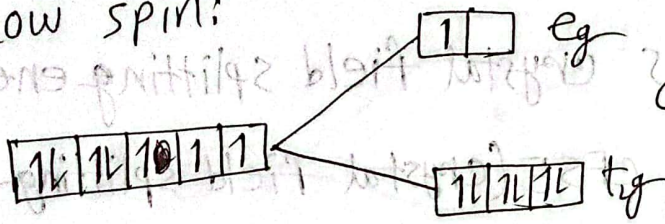
⇒ As high spin:



configuration: $t_{2g}^5 e_g^2$

$\therefore CFSE = [(-0.4 \times 5) + (0.6 \times 2)]\Delta_o = -0.8\Delta_o$

As low spin:



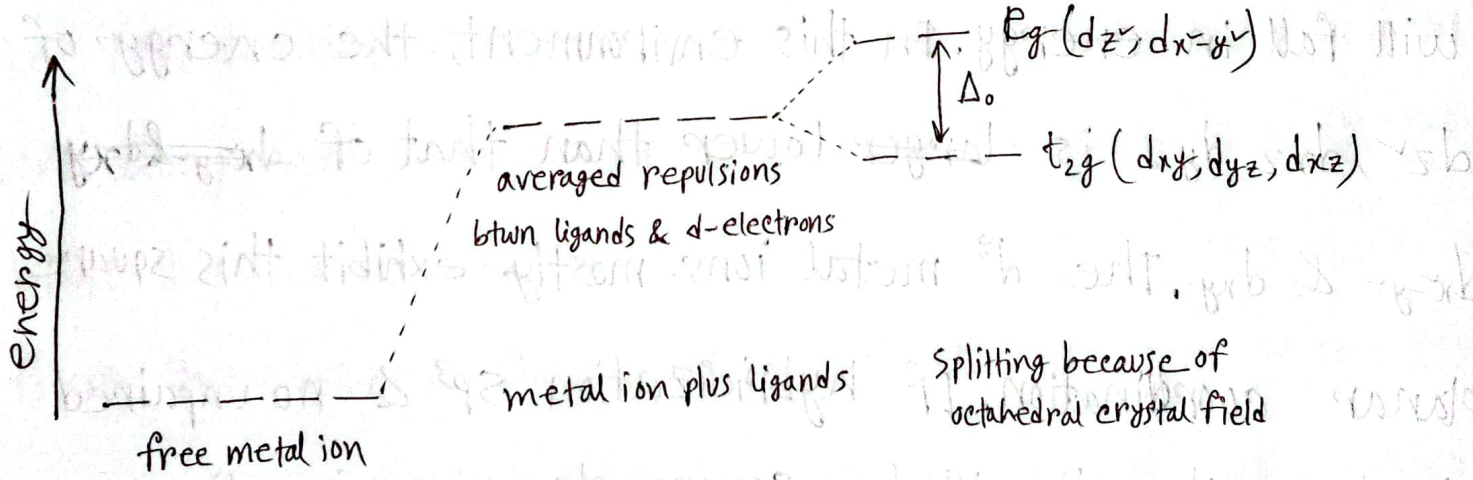
configuration: $t_{2g}^6 e_g^1$

$\therefore CFSE = [(-0.4 \times 6) + (0.6 \times 1)]\Delta_o = -1.8\Delta_o$

⊛ 2018(t), 2017, 2021, 2015(t) Explain the splitting of d-orbitals under the influence of an octahedral field?

⇒ when a transition metal ion placed in octahedral field the surrounding 6 ligands approach the metal along x, y & z axes. this creates an electrostatic interaction btwn the negatively charged ligands & the d-electron ~~& the~~ of metal, which leads to the splitting of d-orbitals. when 6 ligands surrounded the metal ion,

the degeneracy of the d-orbitals is removed. Because two of the orbitals d_{z^2} & $d_{x^2-y^2}$ are directed along the axes pointing the ligands & 3 orbitals d_{xy}, d_{yz}, d_{xz} are directed btwn the axes, away from the ligands. Thus, there is a greater repulsion btwn the e^- of ligands & the $d_{x^2-y^2}$ & d_{z^2} orbitals than d_{xy}, d_{yz}, d_{xz} orbitals. therefore all of the d-orbital are raised in energy because of the electrostatic field generated by the ligands but two of them are raised more than the other three. As a result the d-orbitals are splitted.

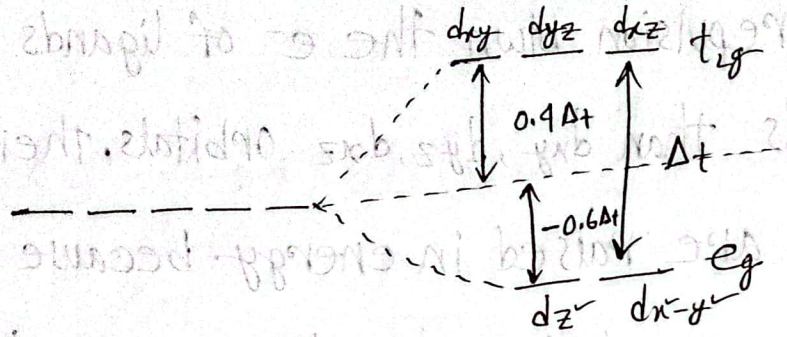


2020, 2016

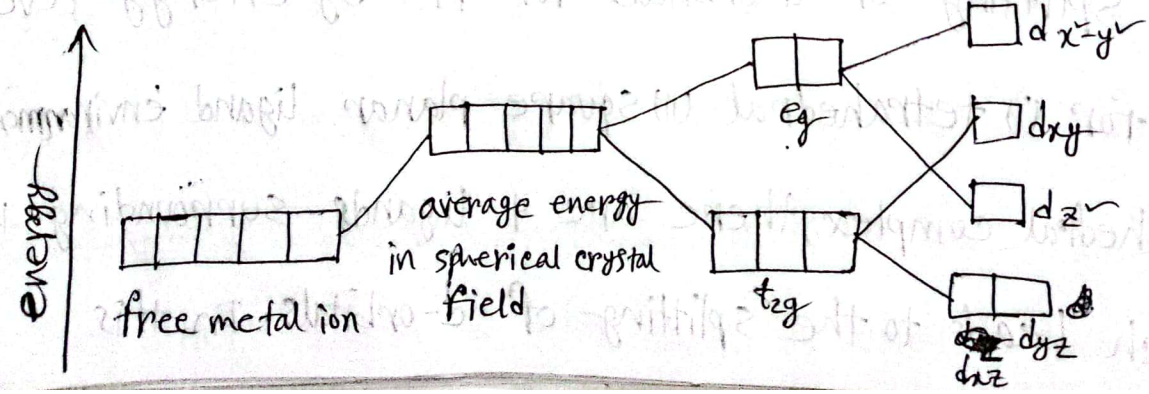
⊗ Show the splitting of d-orbitals for M^{2+} by energy level ~~dig~~ diagram for (i) tetrahedral (ii) square planar ligand environment?

⇒ (i) In tetrahedral complex, there are 4 ligands surrounding the M^{2+} ion, which leads to the splitting of d-orbitals. In this

complex, none of the d-orbitals point directly at the ligands. The d_{xy}, d_{yz}, d_{zx} orbitals have higher energy & are called t_{2g} orbitals & the $d_{z^2}, d_{x^2-y^2}$ orbitals have lower energy & are called e_g orbitals.



(ii) Square planar: In square planar system, there is no ligand along the z-axis & so d-orbitals with a z-compo will fall in energy. In this environment, the energy o d_{z^2}, d_{xz}, d_{yz} is ~~larger~~ lower than that of ~~$d_{x^2-y^2}$ & d_{xy}~~ $d_{x^2-y^2}$ & d_{xy}. The d^8 metal ions mostly exhibit this square planar coordination. If hybridization sp^3 & no unpaired electron then it will be square planar configuration.

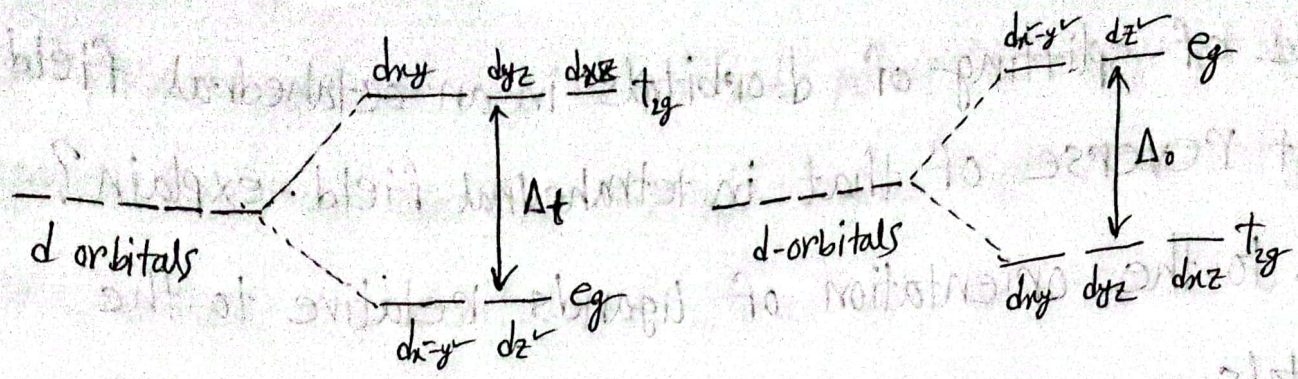


(8) TU Mode of splitting of d-orbitals in an octahedral field is just reverse of that in tetrahedral field - explain?

⇒ Due to the orientation of ligands relative to the d-orbitals.

In octahedral complex, ligands are positioned along x, y & z axes. The d_{z^2} & $d_{x^2-y^2}$ point directly at the ligands & so they experience strong repulsion from the ligands e^- which give them higher energy. The d_{xy}, d_{yz}, d_{xz} are oriented btwn the axes. So they experience less repulsion & are lower in energy.

In tetrahedral complexes, four ligands approach the metal ion along the diagonals of a cube. The d-orbitals that point btwn the axes d_{xy}, d_{yz}, d_{xz} experience greater repulsion from the ligands than the d-orbitals that point along the axes $d_{x^2-y^2}$ & d_{z^2}. This is because ligands are closer to the lobes of the d_{xy}, d_{yz}, d_{xz} orbitals than to the lobes of the $d_{x^2-y^2}$ & d_{z^2} orbitals. So d_{xy}, d_{yz}, d_{xz} experience greater repulsion & have higher energy than $d_{x^2-y^2}$ & d_{z^2}.



for tetrahedral for octahedral

TV

(X) Briefly discuss the factors that affect the crystal field splitting energy (Δ)?

⇒ The most important factors are:

Metal factors	Ligand factors
(I) Oxidation state	(IV) Ligand character
(II) Number of d-electrons	(VI) Number & geometry of ligand
(III) Nature of metal ion	
(IV) Spin pairing energy	

(I) Oxidation state: Higher oxidation state of metal ion causes the ligands to approach more closely to it & the ligands causes more splitting of metal d-orbitals.

(II) The magnitude of Δ decreases with increase in number of d-electrons.

$$\Delta \propto \frac{1}{\text{Number of d-electrons}}$$

(III) Δ increases about 30% to 50% from 3d to 4d & same amount from 4d to 5d.

(IV) Metal ion with higher pairing energy will have lower Δ & vice versa.

(V) strong field ligands cause large splitting & weak field ligands cause for small splitting.

(VI) Δ increase with increase of the number of ligands.

$$\Delta_{sp} > \Delta_{oct} > \Delta_{tet}.$$

2016

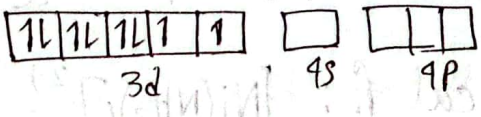
(★) Ni^{2+} react with ligands CN^-, Cl^- & NH_3: one of the complex is octahedral while others are s.p. & tetrahedral:

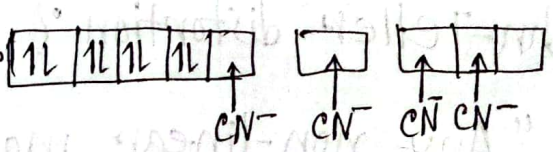
(I) Arrange the ligands to their strength

(II) Assaign ligands to the complexes, give chemical formula?

⇒ (I) $CN^- > NH_3 > Cl^-$

(II) $Ni \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$

$Ni^{2+} \longrightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 \longrightarrow Ar$ 

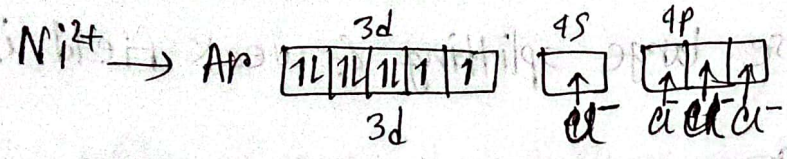
for CN^-: CN^- is strong ligand which can pair the e^- of d-orbitals. $Ni^{2+} \longrightarrow Ar$ 

So the hybridization is dsp^2. And there is no unpaired

electron so CN^- forms square planar (s.p.) geometry with Ni^{2+}

chemical formula: $[Ni(CN)_4]^{2-}$

Cl^-: Cl^- is a weak ligand & cannot pair e^- of d-orbital so it gives high spin configuration.

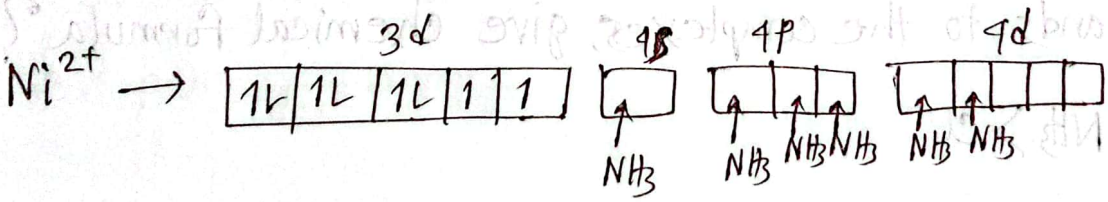


So hybridization is sp^3 & there is two unpaired electron

So it is tetrahedral in geometry.

ci. formular: $[NiCl_4]^{2-}$

for NH_3: N has lone pair e^- & it is neutral which allows 6 ligands to co-ordinate around Ni^{2+}



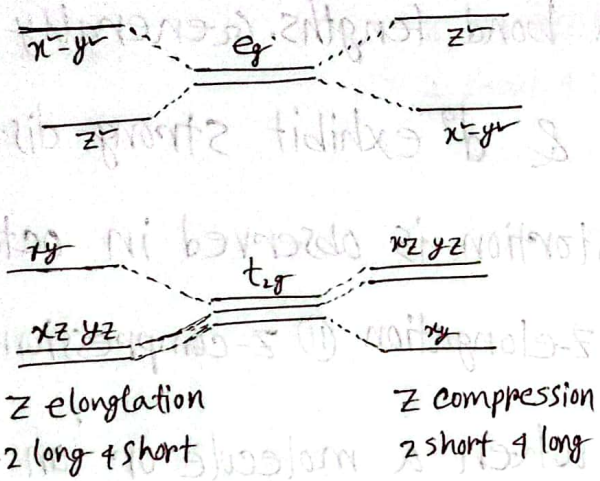
∴ hybridization is sp^3d^2 & shape is octahedral.

chemical f.: $[Ni(NH_3)_6]^{2+}$

⊗ 2016, 2021 What is Jahn-Teller distortion?

⇒ State that "Any non-linear molecular system in a degenerate electronic state will be unstable & will

undergo distortion to form a system of lower symmetry & lower energy thereby removing the degeneracy". It is a principal ~~of~~ that explains the geometrical distortion of certain non-linear molecules & complex ions in degenerate electronic state.



(★) 2021 on the basis of John-Teller theorem explain the distortion of six coordinated complexes?

⇒ The distortion of 6 coordination means octahedral complexes occurs when the complex has a degenerate electronic state in d-orbitals. It arises because the molecule can lower its energy by breaking degeneracy through changes in geometry, such as elongation or compression of bonds. In octahedral complexes, the d-orbitals split into

2 sets:

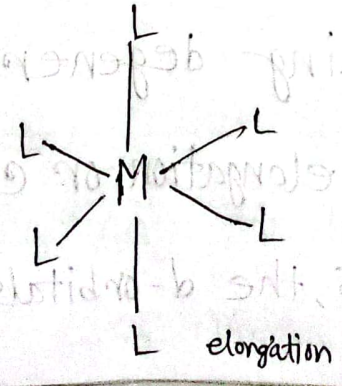
t_{2g} : d_{xy}, d_{yz}, d_{xz} (lower energy)

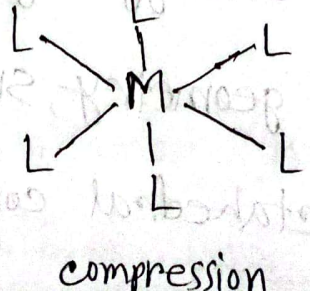
e_g : $d_{x^2-y^2}, d_{z^2}$ (higher energy)

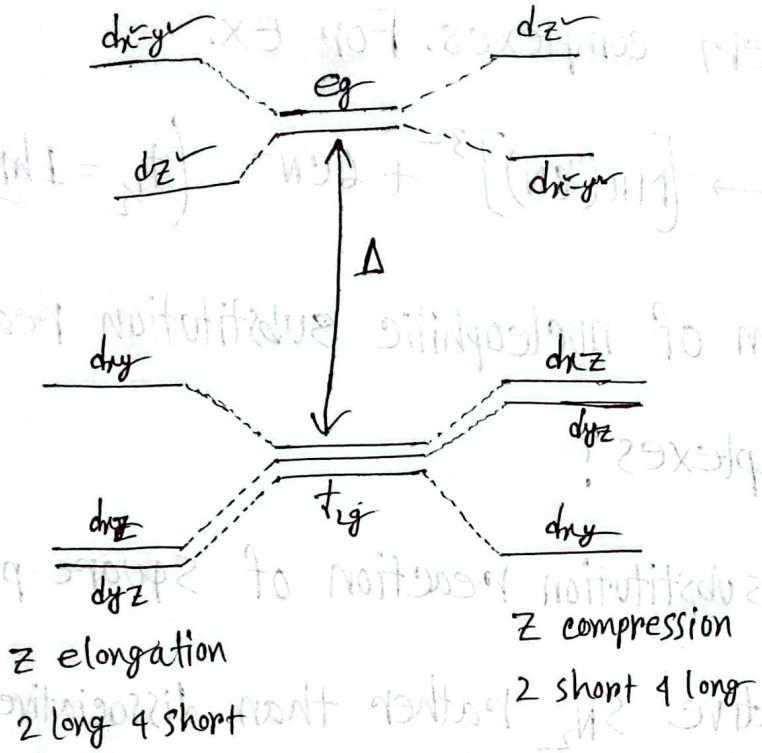
when there is an assymmetric distribution of e^- in t_{2g} or e_g orbitals, it leads to distortion of the orbitals. It results in a change in bond lengths. Generally high spin d^4 & low spin d^7 & d^9 exhibit strong distortion effect. Two types of distortion is observed in octahedral complexes: ~~(I) z elongation~~ (I) z-elongation (II) z-compression.

(I) z-elongation: It occurs when a molecule or ion elongate along the z-axis. It results in a lower of energy those d-orbital that carry a z-component i.e. d_{z^2}, d_{xz}, d_{yz}.

(II) z-compression: It occurs when an octahedral complex compreses along the z-axis. It causes the metals d-orbital with a z-component to become higher in energy.







2021

(X) Explain Lability & Inertness of complexes?

⇒ Lability complexes are those that readily exchange ligands with other species in sln. they have high rate of ligand exchange. In this ~~complexes~~ process, the half life of substitution reaction is 1 min or less than 1 min. For example,

$$[Ni(CN)_4]^{2-} + 4\,{}^{14}CN^- \longrightarrow [Ni({}^{14}CN)_4]^{2-} + 4CN^-$$

there $[Ni(CN)_4]^{2-}$ is stable ion but labile. If add ^{14}C-labelled $^{14}CN^-$ in aq. sln of $[Ni(CN)_4]^{2-}$, CN^- ligands are replaced by $^{14}CN^-$.

⇒ Inert complexes: the complexes in which ligand substitution occurs by the other ligand slowly & whose half life is more

than 1min is called inert complexes. For Ex.

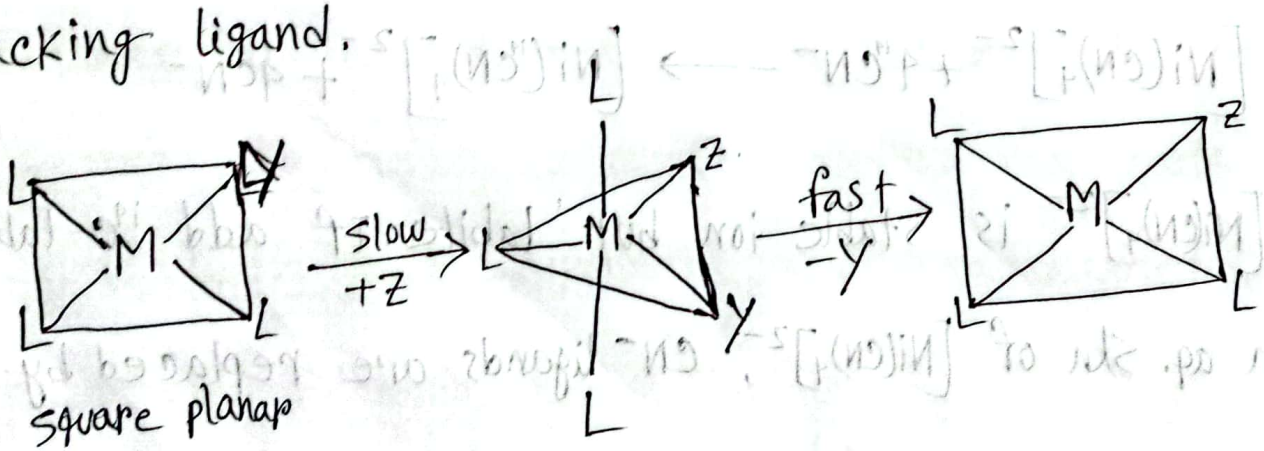
$$[Mn(CN)_6]^{3-} + 6{}^{11}CN^- \longrightarrow [Mn({}^{11}CN)]^{3-} + 6CN^- \quad (t_{1/2} = 1\,hr)$$

2021

Discuss the mechanism of nucleophilic substitution reaction in square planar complexes?

⇒ The mechanism of substitution reaction of square planar appears to be associative S_{N_2} rather than dissociative S_{N1}.

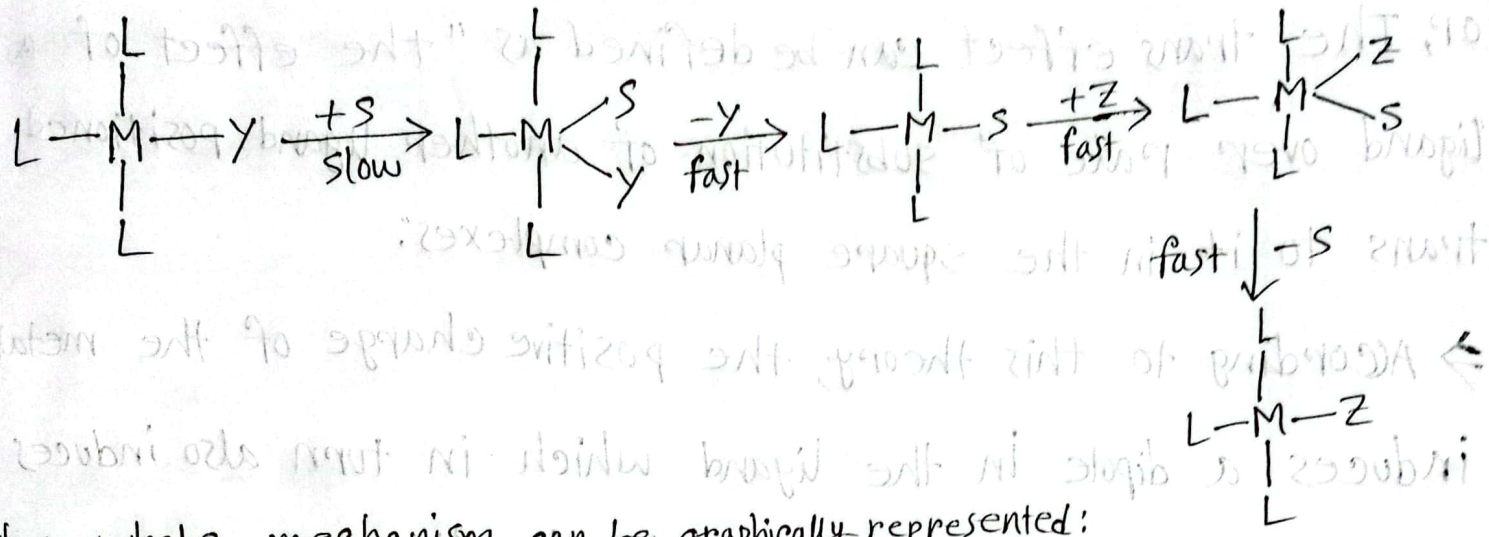
In case of square planar complexes of Ni(II), Pt(II) five empty metal orbital (one d, one s & three p) of comparable energies can be made available for sigma bonding. Since only four ligands are to be σ-bonded with metal ion only. The fifth orbital can easily accommodate e⁻ from the attacking ligand.



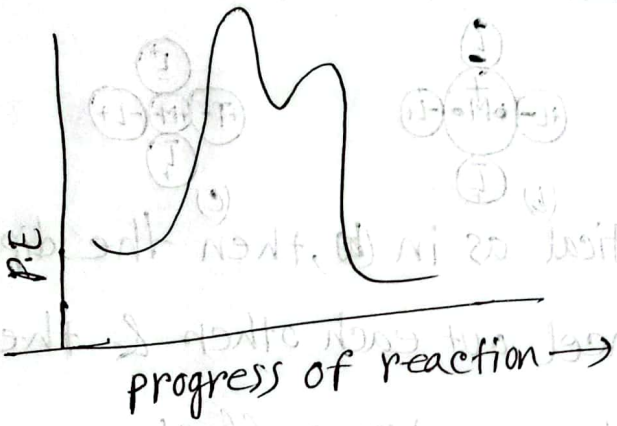
reaction: $ML_3Y + Z \longrightarrow ML_3Z + Y$

r. rate $= K[ML_3Y]^1[Z]^1$ & order = 2.

when it is associated with Solvent (s), then the mechanism:



The whole mechanism can be graphically represented:



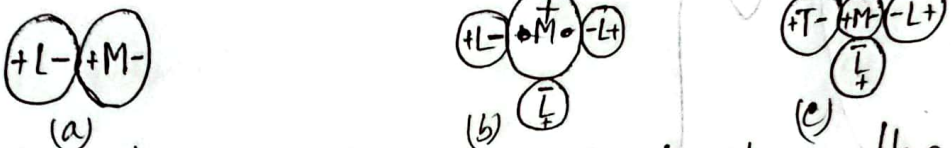
(2021) What is trans effect? Explain it in term of electrostatic polarization theory?

⇒ It is observed that during the substitution reaction of square planar metal complexes, some ligands preferentially direct the substitution trans to themselves. such ligands are called trans directing & tendency of these ligand to direct the incoming ligand to occupy the position trans to it, is called

trans effect.

or, The trans effect can be defined as "the effect of a ligand over rate of substitution of another ligand positioned trans to it in the square planar complexes".

⇒ According to this theory, the positive charge of the metal induces a dipole in the ligand which in turn also induces a dipole in the metal as shown in the figure:

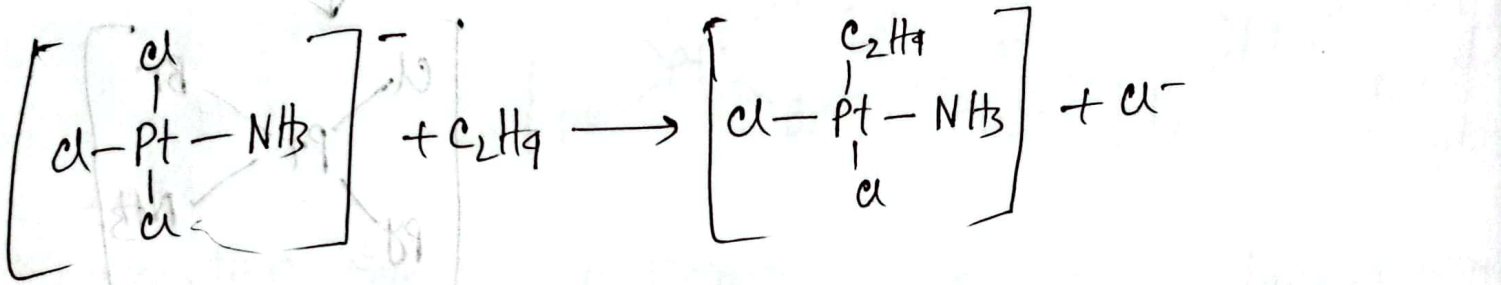


Now if all 4 ligands are identical as in (b), then the dipoles induced by the metal ion cancel out each other & the resultant dipole is zero. Hence they show no trans effect.

But if the 4 ligands are not identical as in (c), then the induced dipole do not cancel out each other. The 2 L ligand which are similar & trans to e/o will balance e/o (each other). But the 2 trans ligands, T & L will whave an effect on each other. Here, T can polarise M more than the trans L. for this, there will be a decrease in +ve charge on the side trans to T. This results in an increased bond length of M-L

& so the M-L bond becomes weak. Hence substitution of ligand, L becomes easier on the side trans to T. that is, T here acts as a trans directing group.

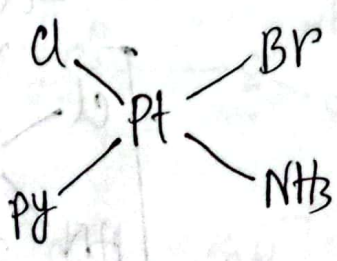
⇒ diagram for ~~A~~ trans effect:



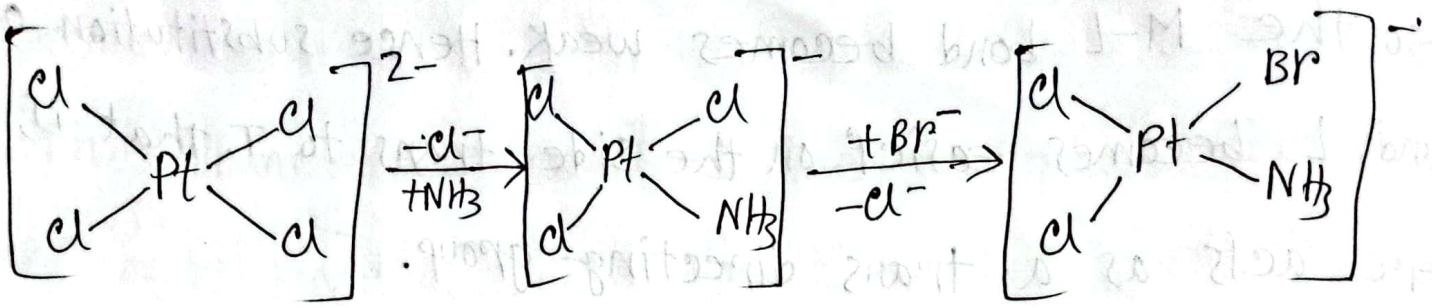
Cl^- has greater trans effect than NH_3, so trans of Cl^- replaced by C_2H_4. Cl^- trans Cl^- is replaced but not the Cl^- which trans to NH_3.

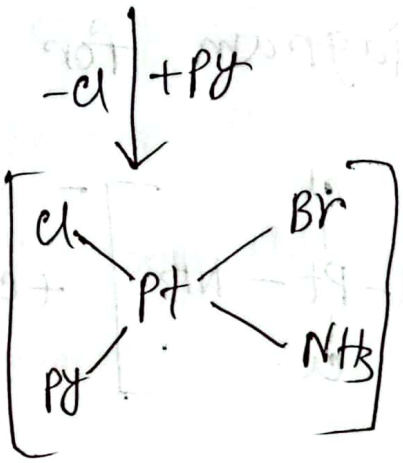
2024(t)

(X) Using $K_2[PtCl_4]$ and other appropriate reagents, show by reaction scheme how will you prepare the compound (Given trans effect $Br^- > Cl^-$. $py > NH_3$)?



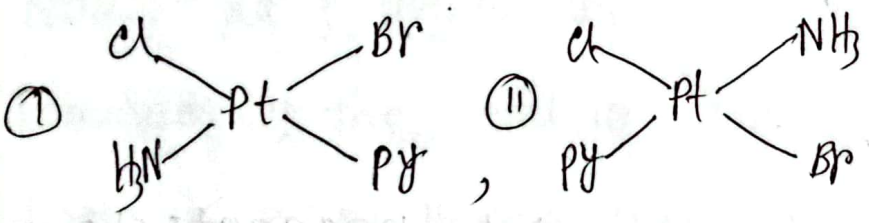
⇒ $K_2[PtCl_4]$ contain the metal ion with ligand $[PtCl_4]^{2-}$.



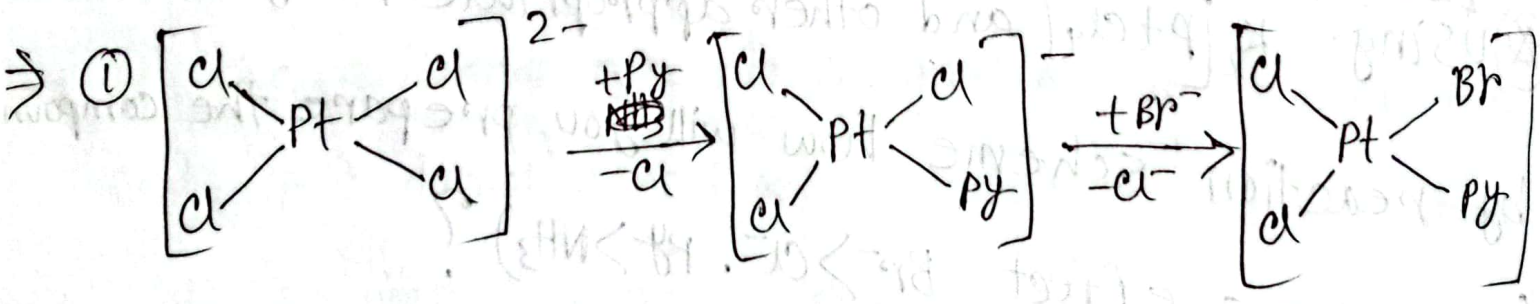


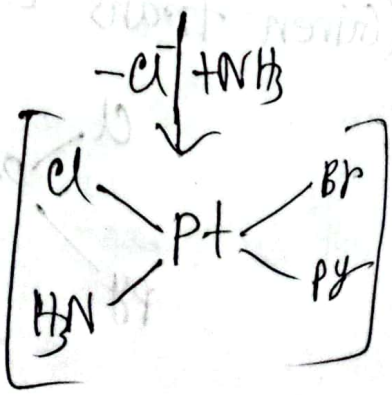
Extra

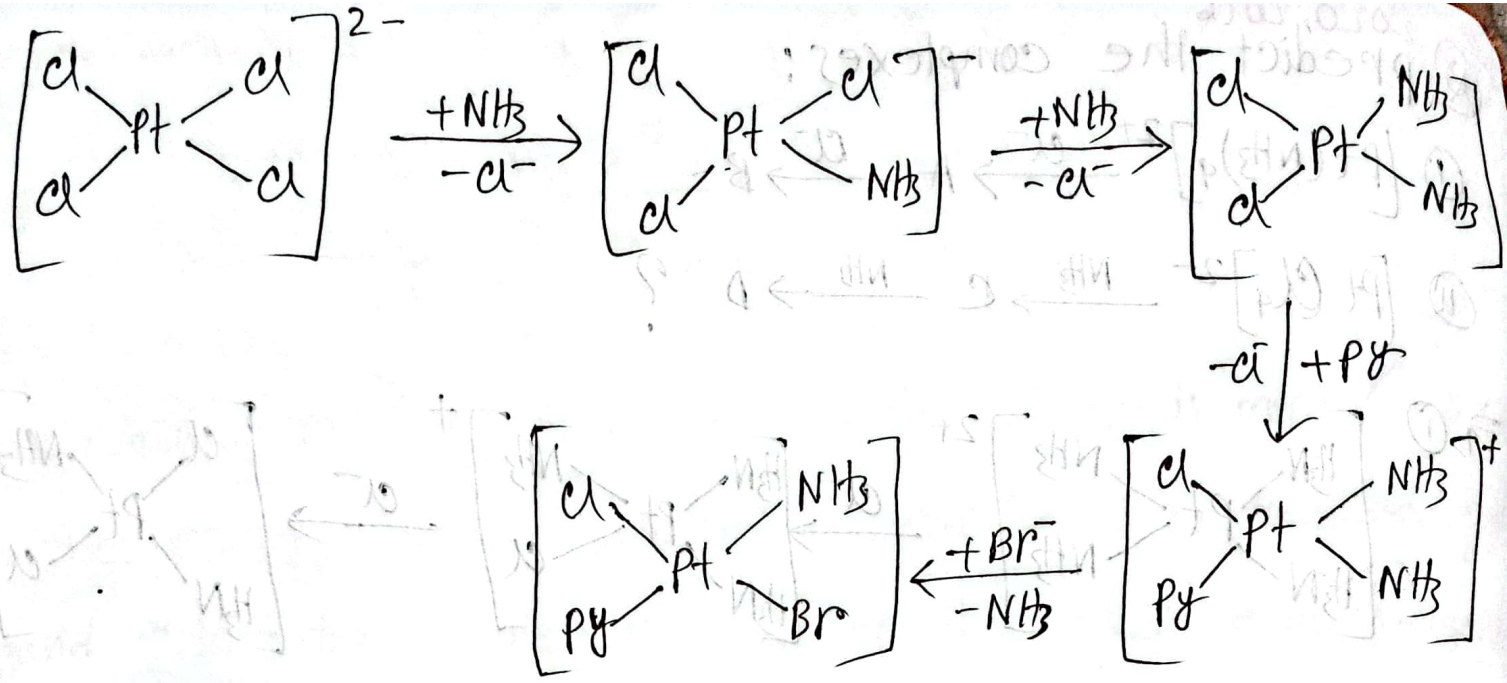
✗ prepare the following compound :



⇒





(11) 

2021, 2022

(12) Show the substitution reaction mechanism in $[Co(NH_3)_5CO_3]^+$ to produce $[Co(NH_3)_5(H_2O)]^{3+}$ with out breaking metal-ligand bond?

$\Rightarrow [Co(NH_3)_5(CO_3)]^+ + 2H_3O^+ \longrightarrow [Co(NH_3)_5(H_2O)]^{3+} + CO_2 + 2H_2O$

If we use ^{18}O labelled H_2O^{18} we can see that all oxyge

of $[Co(NH_3)_5(H_2O)]^{3+}$ complex come from CO_3^{2-} group.

$[(NH_3)_5 Co^{III}-\ddot{O}-CO_2]^+ + H_3O^+$

↓

$[(NH_3)_5 Co^{III}-\ddot{O}(H^+)(CO_2)]^{2+} + H_2O$

↓

$[(NH_3)_5 Co^{III}-\ddot{O}H]^{2+} + CO_2$

↓

$[(NH_3)_5 Co^{III}\leftarrow \ddot{O}(H)(H)]^{3+} + H_2O$

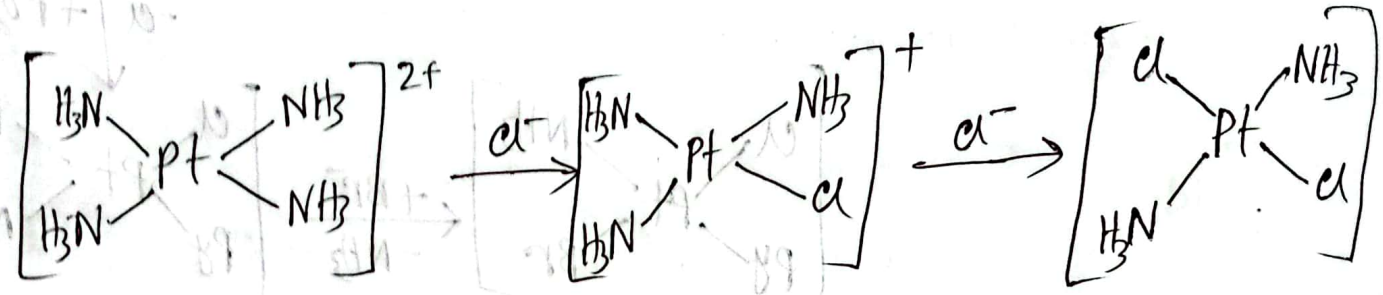
2020, 2016

Predict the complexes:

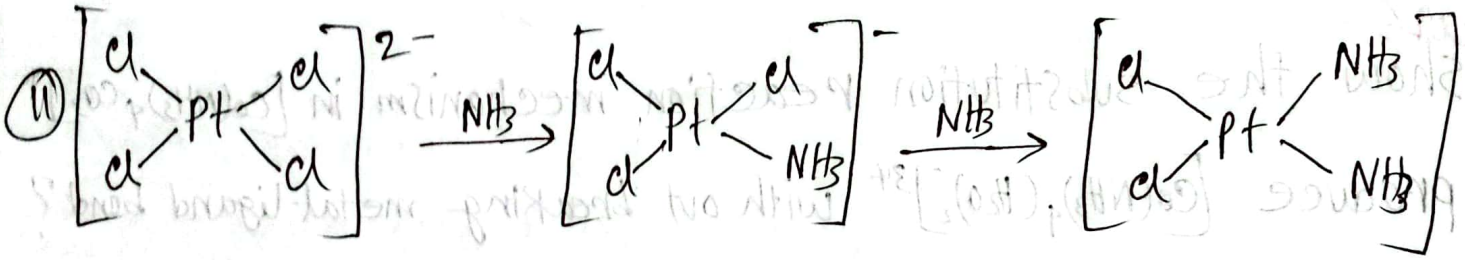
(i) $[Pt(NH_3)_4]^{2+} \xrightarrow{Cl^-} A \xrightarrow{Cl^-} B$

(ii) $[PtCl_4]^{2-} \xrightarrow{NH_3} C \xrightarrow{NH_3} D$?

⇒ (i)



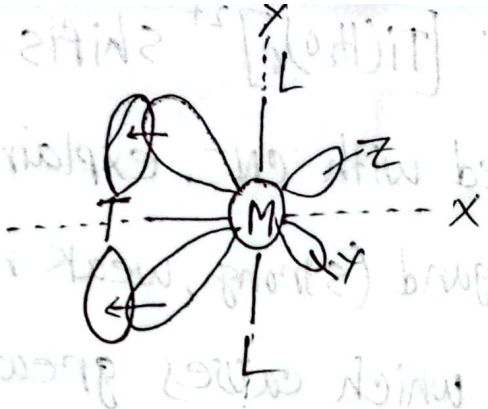
(ii)



2020

Discuss the π-bonding theory proposed ~~by~~ for trans effect?

⇒ It explains the phenomenon in term of electronic interactions btwn metal & ligands. This theory is applicable for π-acceptor ligands only such as CN^-, NO, CO, C_2H_4 etc. These ligands have empty π or π* orbital that can accept e⁻ density from metals d-orbitals. It reduces the e⁻ density on the metal on the side trans to the π-acceptor ligand. It contributes to the formation of a five coordinate complex with the leaving & incoming group in it.



5-coordinate transition state formed in this way is more stabilized & the energy of the substitution reaction. Thus the rate of ligand substitution is increased due to the presence of π-bonding trans ligands.

2020, 2019, 2017

(X) Cr^{2+} ion easily oxidized to Cr^{3+} - explain by CFT ?

⇒ because Cr^{2+} becomes more stable when it oxidizes to Cr^{3+}.

$Cr^{2+} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^0$

$Cr^{3+} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$

In Cr^{2+} the 4-electron are arranged as $t_{2g}^3 e_g^1$, while the 3 d-electron arraged as $t_{2g}^3 e_g^0$ in Cr^{3+}. That means the t_{2g} orbitals are half filled in Cr^{3+} & so it is more stable. Thus Cr^{2+} readily loses an e^- to achieve the more stable t_{2g}^3 configuration & gets oxidized.

(*) 2018 The maximum absorption of $[Ti(H_2O)_6]^{2+}$ shifts from 500 nm to 450 nm when H_2O is replaced with CN^-. explain?

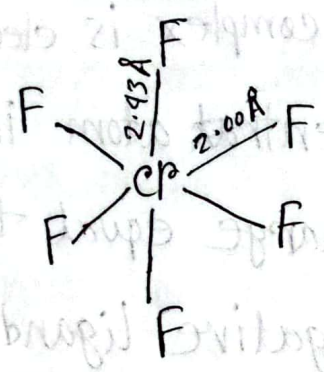
⇒ Due to nature of the ligand (strong, weak nature) CN^- is strong field ligand which causes greater splitting energy than H_2O. As a result transition energy also becomes higher. Energy is inversely proportional to wavelength $[E = \frac{hc}{\lambda}]$ so when H_2O is replaced by CN^- the Δ will increase & λ will decrease from 500 nm to 450nm.

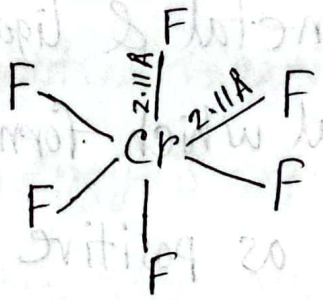
(*) $[CrF_6]^{4-}$ & $[MnF_6]^{4-}$ both have 6 F^- ligand but Mn-F bond length are equivalent but Cr-F are not - why?

⇒ $Cr^{2+} \rightarrow 1s^2 2s^2 2p^6 3s^2 3d^4 \longrightarrow t_{2g}^3 e_g^1$

$Mn^{2+} \rightarrow 1s^2 2s^2 2p^6 3s^2 3d^5 \longrightarrow t_{2g}^3 e_g^2$

Mn^{2+} has a symmetrical distortion of e^- in both t_{2g} & e_g orbitals. Hence no Jahn-Teller distortion occurs & so all the Mn-F bonds are equivalent. But in Cr^{2+}, the e^- are not symmetrically distributed in e_g-orbital, which leads to Jahn-teller distorsion. The complex elongated or compresses along z-axis causing unequal Cr-F bond length.





(A) 2019 which complex is high spin in the pair?

① $[Cr(H_2O)_6]^{2+}$ & $[Cr(CN)_6]^{4-}$ ⑪ $[Fe(H_2O)_6]^{2+}$ & $[Fe(CN)_6]^{4-}$

⇒① As CN^- is strong ligand so $[Cr(CN)_6]^{4-}$ is low spin & $[Cr(H_2O)_6]^{2+}$ is high spin.

⑪ CN^- is strong ligand but H_2O is weak ligand so $[Fe(H_2O)_6]^{2+}$ is high spin & $[Fe(CN)_6]^{4-}$ is low spin.

(A) 2019 Describe CFT for transition metal complexes?

⇒ "A co-ordinating group or ligand when approach, the orbital of central metal ion, then influence the charge cloud of the metal ion which provides an electrostatic field" known as crystal field.

The theory that deals with the environmental electrostatic field know as CFT.

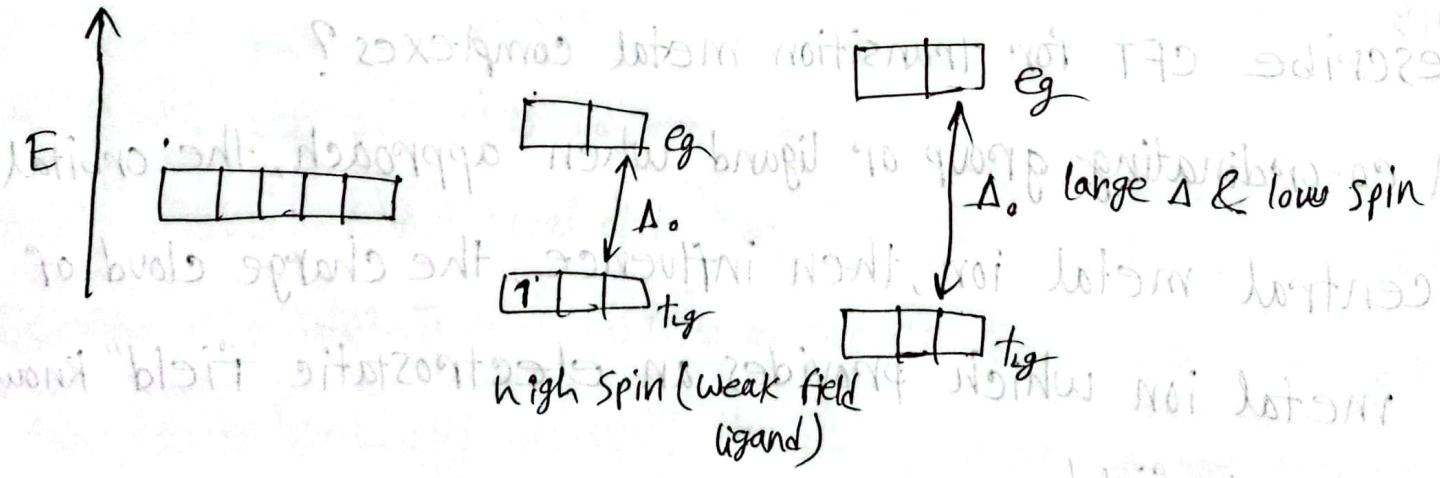
CFT for transition metal complexes assumes that the attraction

btwn the central metal & ligands in a complex is electrostatic. The transition metal which forms the central atom in complex is regurded as positive ion of charge equal to the oxidation state. This is surrounded by negative ligand or neutral molecules which have lone pairs of e⁻. some assumptions of CFT:

① ligands are treated as point charges.

② The d-orbitals of all the metals have same energies.

When complex is formed, the ligands destroy the degeneracy of these orbitals, the orbitals now have different energies. The d-orbitals split into two groups: t_{2g}, e_g.



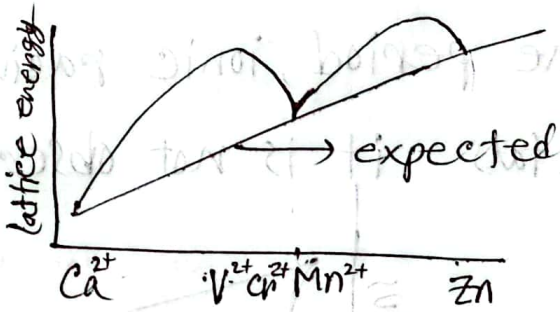
⊗ 2021 Thermodynamic properties of transition metal complexes are affected by d-orbital splitting - explain?

⇒ ① Lattice energy: It is the energy required to break

apart an ionic solid & converts its constituent atoms into gaseous ions. The lattice energy is expected to increase continuously across the transition series as we go from left to right due to the decrease in ionic radius of metal ions. But as we expected, a smooth curve is not seen, instead we get a double hump shaped curve. this is because of the CFSE value of the metals. the values are

Metals:	Ca^{2+}	Sc^{2+}	Ti^{2+}	V^{2+}	Cr^{2+}	Mn^{2+}	Fe^{2+}	Co^{2+}	Ni^{2+}	Cu^{2+}	Zn^{2+}
confi:	d^0	d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
CFSE, Δ_o:	0	−0.4	−0.8	−1.2	−0.6	0	−0.4	−8	−1.2	−0.6	0

The CFSE value of Ca^{2+}, Mn^{2+} & Zn^{2+} is zero (0). Hence they follow expected line. But with the increase & ~~Late~~ decrease of the value, deviations of lattice energy from the expected line is observed. Hence we get a double humped shaped graph.

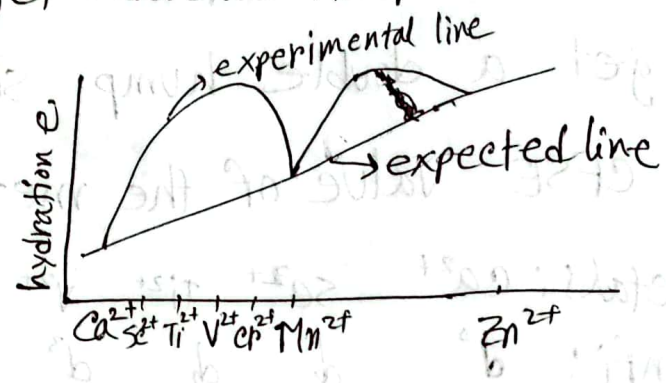


② Heat of hydration: It is the energy released when new bonds btwn metal ions & water molecules are made. it is also expected to increase smoothly from Left to right of the transition metals. But due to CFSE value we do not

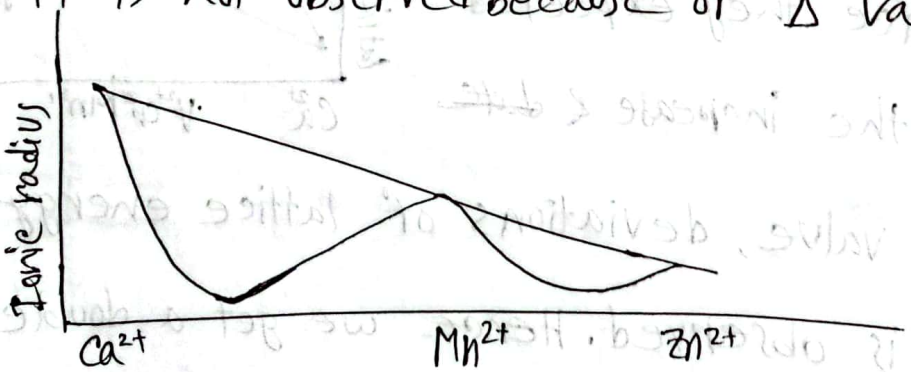
observe a smooth increase in value.

$M^{2+} + 6H_2O \longrightarrow [M(H_2O)_6]^{2+}_{(aq)}$ – heat of hydration

the hydration energy for Sc^{2+}, Ti^{2+}, V^{2+} is increased & decreased for Cr^{2+}; same kind of increase observed for $Fe^{2+}, Co^{2+}, Ni^{2+}$ & Cu^{2+}. so we get double humped shape (graph again.



③ Ionic radius: usually moving from the left to right in same period, ionic radius of ions decreases. But in transition metals it is not observed because of Δ value.



④ what are the difference btwn: splitting in octahedral & tetrahedral fields?

⇒ ① There are 4 ligands in tetrahedral field but there

are six ligands present in octahedral.

(II) None of the d-orbitals point directly at the ligands ~~present~~ in a tetrahedral field while two d-orbitals point directly at the ligands in octahedral field.

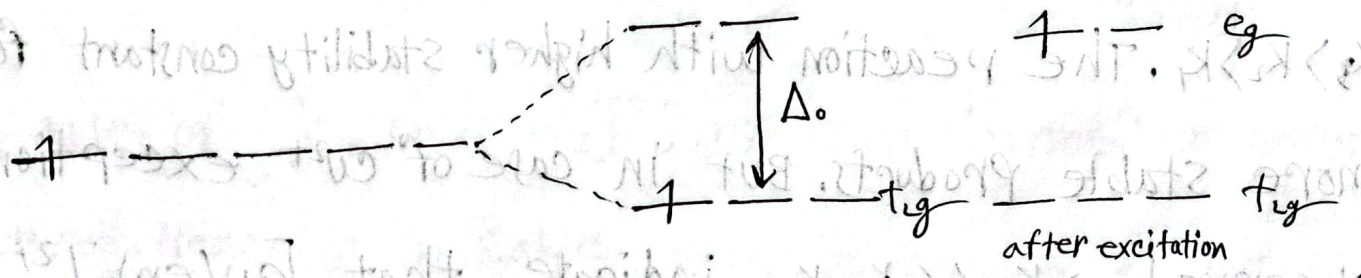
(III) Δ_t smaller than Δ_o.

(IV) Tetrahedral field usually exhibit high spin because Δ_t is small while octahedral can exhibit both high & low spin depending on ligands.

(V) In tetrahedral field, e_g orbitals have lower energy than t_{2g} while in octahedral e_g orbitals have higher energy than e_g.

(*) Why is Ti^{3+} ion soln is violet in colour?

⇒ When Ti^{3+} is surrounded by six ligands, it forms an octahedral complex. The 3d orbital of Ti^{3+} split into two sets due to the crystal field ~~due to the~~ by the ligands.



Since Ti^{3+} has one e^- in 3d orbital, it will initially reside

in t_{2g} orbital. When light strikes the soln, e^- absorbs energy & excites from t_{2g} to e_g level. In case Ti^{3+}, it absorbs green/yellow/red photons. The remaining transmitted light appears as violet which we see. Hence the soln of Ti^{3+} is violet.

⊗ 2021 Ethylene diammine forms stable chelate with Cu^{2+} yet $[Cu(en)_3]^{2+}$ ion is highly unstable - explain?

⇒ For any metal (II) ions, the formation of chelate complex with "en" by elimination of H_2O includes following steps:

$$[M(H_2O)_6]^{2+} + en \xrightarrow{K_1} [M(en)(H_2O)_4]^{2+}$$

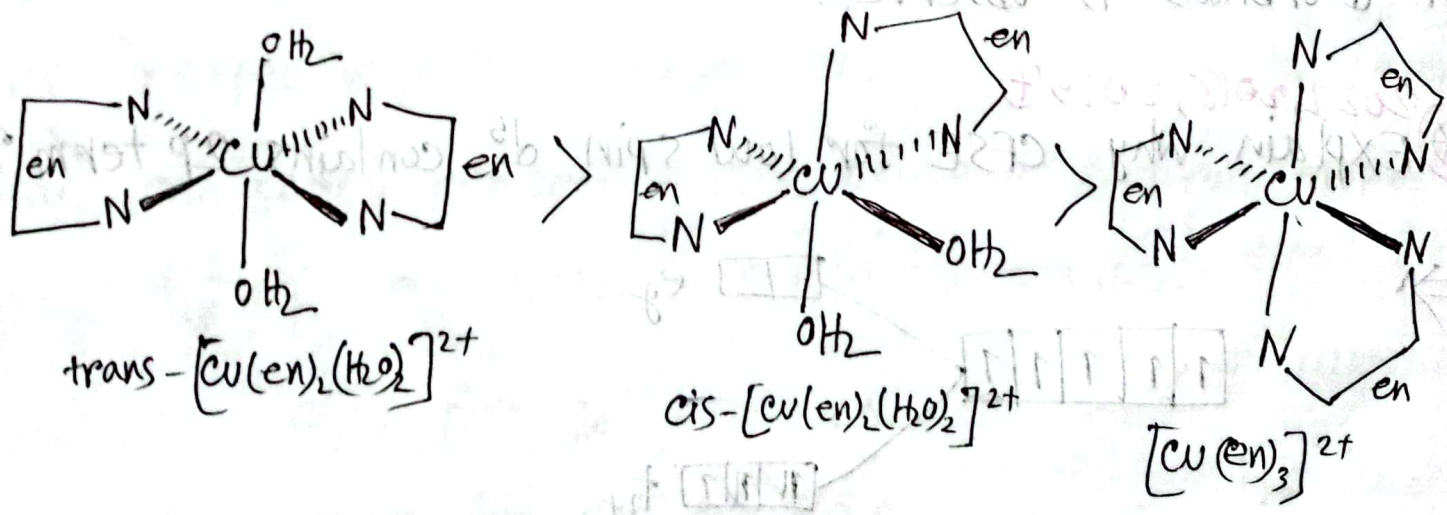
$$[M(en)(H_2O)_4]^{2+} + en \xrightarrow{K_2} [M(en)_2(H_2O)_2]^{2+}$$

$$[M(en)_2(H_2O)]^{2+} + en \xrightarrow{K_3} [M(en)_3]^{2+}$$

For any metal, the stability ~~order~~ constant order is $K_3 > K_2 > K_1$. The reaction with higher stability constant forms more stable products. But in case of Cu^{2+} exception is observed → $K_3 << K_1, K_2$ indicate that $[Cu(en)_3]^{2+}$ is

highly unstable. This unability is due to Jahn-teller distortion. It states that in a degenerate electronic state a non-linear molecule will undergo a distortion to remove degeneracy & stabilize the system. For d^9 systems like Cu^{2+} this distortion is highly significant.

In $[Cu(en)_3]^{2+}$, Cu^{2+} will experience distortion along z-axis. "en" is a bidentate ligand forming 3 chelate rings around the Cu center. The Jahn-teller effect distorts the octahedral geometry causing a strain in the chelate rings due to the strain into "en" molecule, $[Cu(en)_3]^{2+}$ is highly unstable.



But in case of $[Cu(en)_2(H_2O)_2]^{2+}$, no strain in the "en" molecule is observed. Hence it is stable.

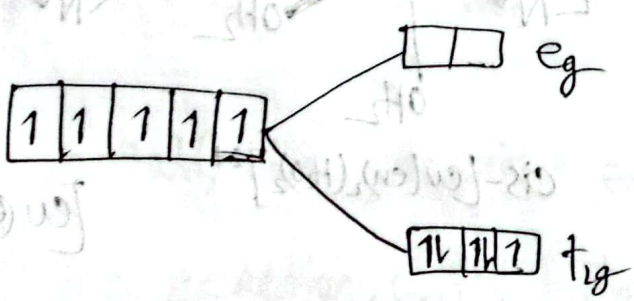
⊛ Why is the splitting energy of d-orbitals in tetrahedral complex smaller than octahedral?

⇒ ① In tetrahedral, there are only 4 ligands while octahedral has 6. This fewer number of ligands reduces the overall ligand field strength leading to a smaller splitting of the d-orbitals.

② The ligands in tetrahedral interact less directly with the d-orbitals because none of d-orbitals point directly at the ligands. The ligands are positioned btwn the axes which reduces the interaction btwn ligands & d-orbitals. So weaker splitting of d-orbitals is observed.

⊛ 2021, 2018, 2019(t)
Explain Why CFSE for low spin d^5 contains 2P term?

⇒



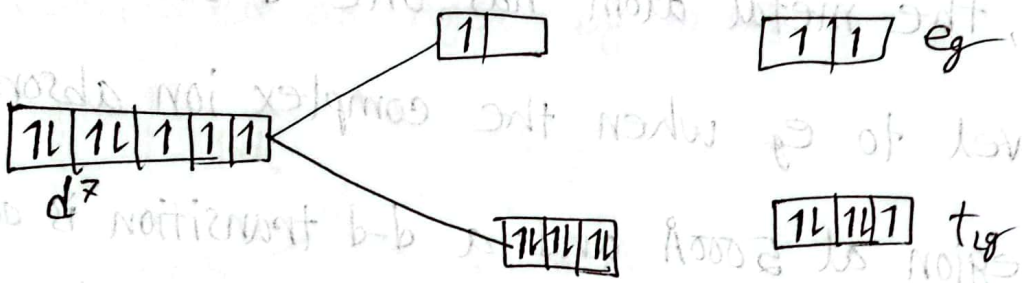
there are two ~~fone~~ pair electrons in t_{2g} set, each incurring a pairing energy cost of P. So total pairing energy contribution is 2P.

∴ Total CFSE = $(-0.4 \times 5 + 0.6 \times 0)\Delta_o + 2P$

$= -2\Delta_o + 2P$.

⑧ Co(II) is stable in aquous ~~sta~~ medium but not stable in NH_3 ?

⇒ Co(II) ion posses d^7 system, has two possible configuration $t_{2g}^6 e_g^1$ & $t_{2g}^5 e_g^2$



Since H_2O weak & NH_3 strong ligand high spin & low spin occurs respectively. Thus single electron in low spin in e_g orbital can easily removed & can oxidized & form more stable $t_{2g}^6 e_g^0$. So Co(II) is stable in ~~NH_3~~ H_2O.

CFSE for $Co^{3+} = -2.4\Delta_o$

CFSE for $Co^{2+} = -1.8\Delta_o$

⑧ $[Co(NH_3)_6]^{2+}$ is a paramagnetic ion which is readily oxidized to give diamagnetic ion $[Co(NH_3)_6]^{3+}$, why?

⇒ Co^{2+} ion has d^7 configuration. But Co^{3+} has d^6 configuration

in $[Co(NH_3)_6]^{3+}$ results in a half filled t_{2g} set, which is a stable electronic configuration. oxidation of Cu^{2+} removes ($t_{2g}^6 e_g^1$) the unpaired e^- in the e_g orbital, leading to a more stable diamagnetic complex.

2019, 2017
Explain why $[Ti(H_2O)_6]^{3+}$ is violet but $TiCl_3$ is colourless?

⇒ In $[Ti(H_2O)_6]^{3+}$, the metal atom has one 'd' electron which undergo t_{2g} to e_g level (transition) when the complex ion absorbs light in the visible region at 5000Å and a d-d transition is occured. d^1 e^- can be promoted from t_{2g} to the e_g level by absorbing visible light. The λ of light absorbed a specific energy & complementary color (violet) is observed.

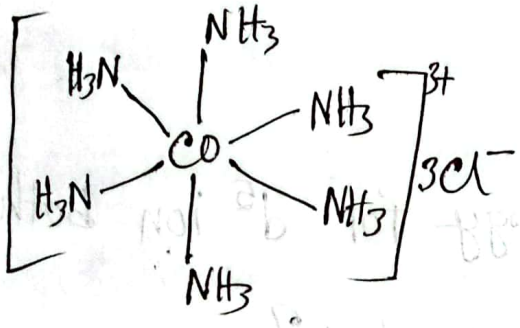
In Solid $TiCl_3$, Ti^{3+} ions are not surrounded by ligands. This means there is no ligand field to split the d-orbitals. without the splitting, the d-electron cannot absorbed specific λ of light to undergo a d-d transition. As no light is absorbed in the visible range, $TiCl_3$ appears colorless.

Rajib
che-50

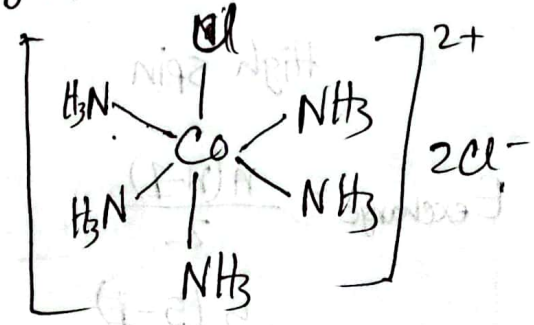
CamScanner

(xx) Formulate $CoCl_3 \cdot 6NH_3$, $CoCl_3 \cdot 5NH_3$, $CoCl_3 \cdot 4NH_3$ & $CoCl_3 \cdot 3NH_3$ by warner theory if 3, 2, 1 & chloride ions are precipitated as agcl from their Aq. Soln ?

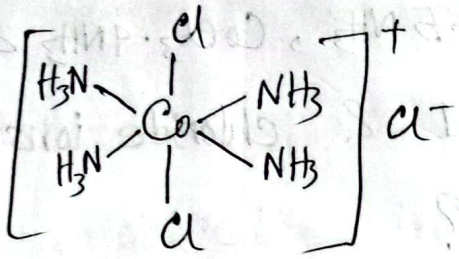
⇒ warner deduced that in $CoCl_3 \cdot 6NH_3$ three Cl acted as primary valancies & six NH_3 as secondary valancies & the complex is written as $[Co(NH_3)_6]Cl_3$. The three Cl^- ar ionic & precipitated as AgCl by $AgNO_3$.



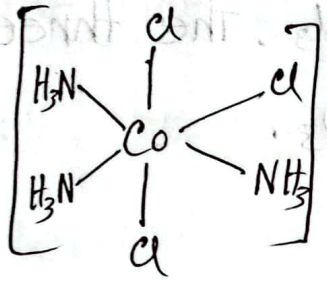
$CoCl_3 \cdot 5NH_3$ written as $[Co(NH_3)_5Cl]Cl_2$. It has two primary (two Cl) and six secondary valancies (5 NH_3 & on Cl). Two Cl^- are precipitated by $AgNO_3$.



$CoCl_3 \cdot 4NH_3$ has ~~two~~ one primary (one Cl^-) and six secondary valancies (4 NH_3 & 2 Cl^-) which is written as $[Co(NH_3)_4Cl_2]Cl$.



$CoCl_3 \cdot 3NH_3$ has no primary valancies but has six secondary valancies. It is written as $[Co(NH_3)_3Cl_3]$.



2018, 2019 (t)

Calculate exchange energy for d^5 ion both as a high & low spin in an octahedral complex?

$\Rightarrow$

$\underline{1}\ \underline{1}\ \underline{1}\ \underline{1}\ \underline{1}$

d^5, Degenerate

$\underline{1}\ \underline{1}\ \underline{1}\ \underline{1\downarrow}\ \underline{1\downarrow}\ \underline{1}$

High spin | low spin

$E_{exchange} = \frac{n(n-1)}{2}$

$= \frac{5(5-1)}{2}$

$= 10$

$E_{exchange} = \frac{n(n-1)}{2}$

$= \frac{3(3-1)}{2} + \frac{2(2-1)}{2}$

$= 3+1$

$= 4$

$\therefore$ Loss of exchange energy $= 10-4 = 6$

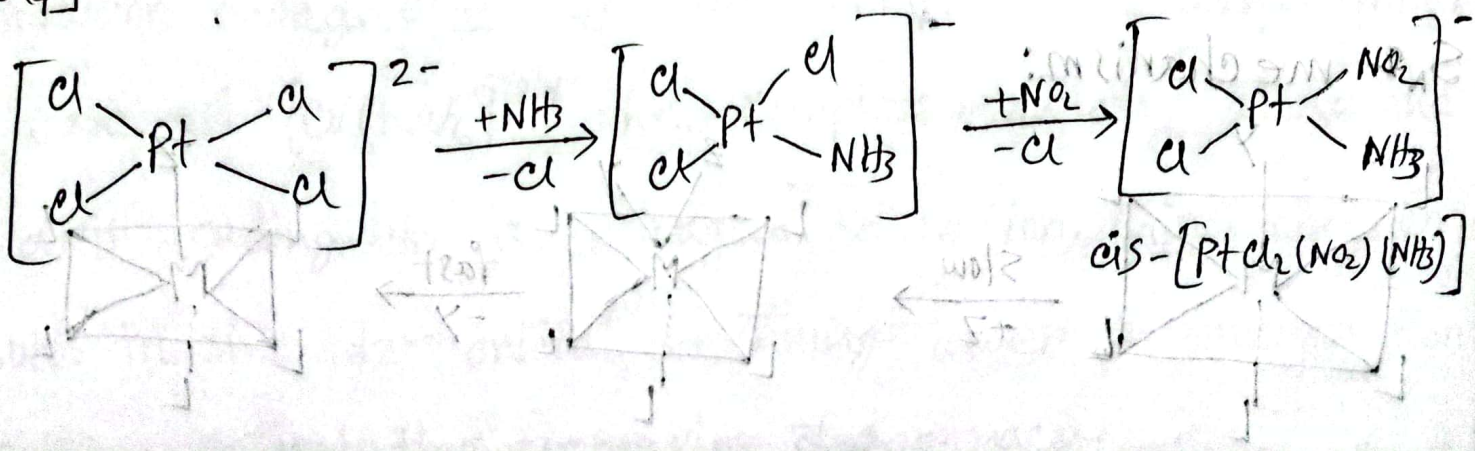
2020, 2019, 2020(t)

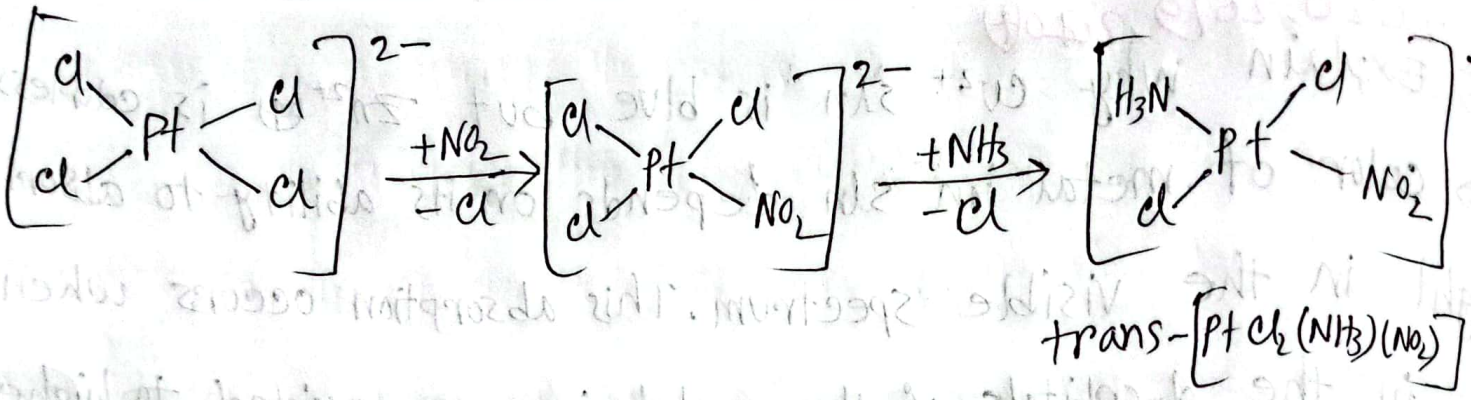
(A) Explain why Cu^{2+} soln is blue but Zn^{2+} (aq) is colorless?

⇒ color of metal ion soln depends on its ability to absorb light in the visible spectrum. This absorption occurs when e^- in the d-orbitals of the metal ion are excited to higher energy level. Cu^{2+} ion have d^9 means incomplete d-orbital, with one unpaired e^-. This allows d-d transitions meaning e^- can absorb specific λ of light & jump to higher energy levels. Cu^{2+} soln absorbs light in the red-orange region, making the soln appear blue.

But Zn^{2+} ions have d^{10} means filled d-orbital. so no d-d transition occurs. thats why Zn^{2+} ions cannot absorb ~~light~~ visible light, making their soln appear colorless.

(A) Design synthesis of cis & trans $[PtCl_2(NH_3)(NO_2)]$ from $[PtCl_4]^{2-}$?

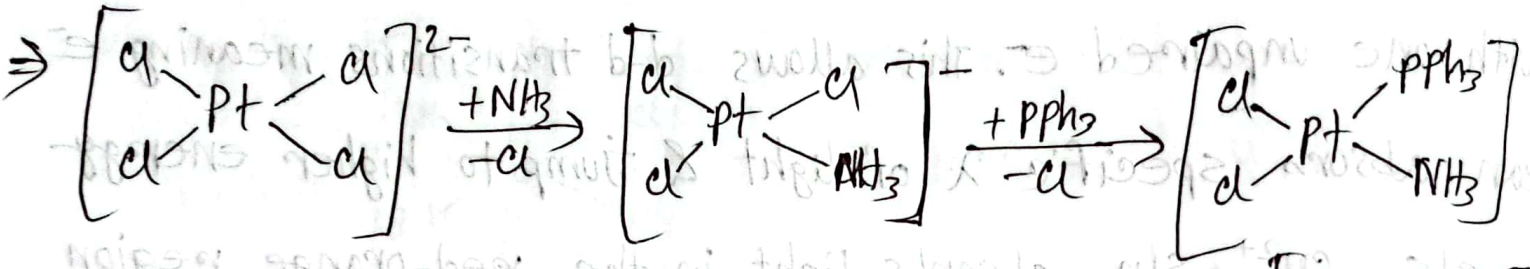
⇒ 



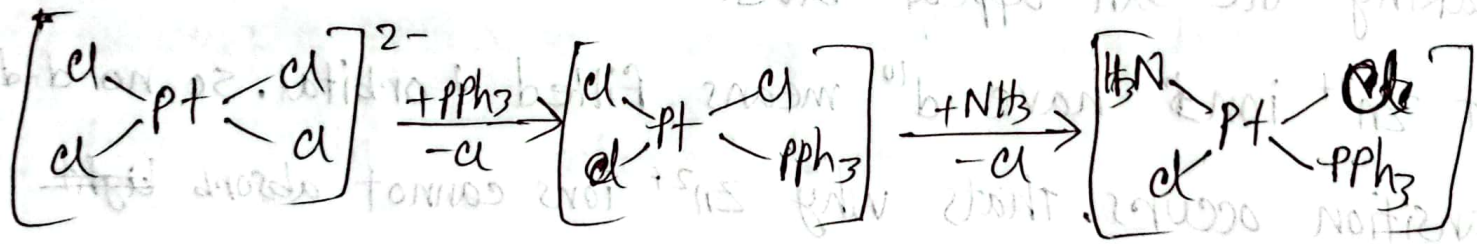
trans-$[PtCl_2(NH_3)(NO_2)]$

extra

Write reaction to produce cis & trans-$[PtCl_2(NH_3)(PPh_3)]$?



cis-$[PtCl_2(NH_3)(PPh_3)]$



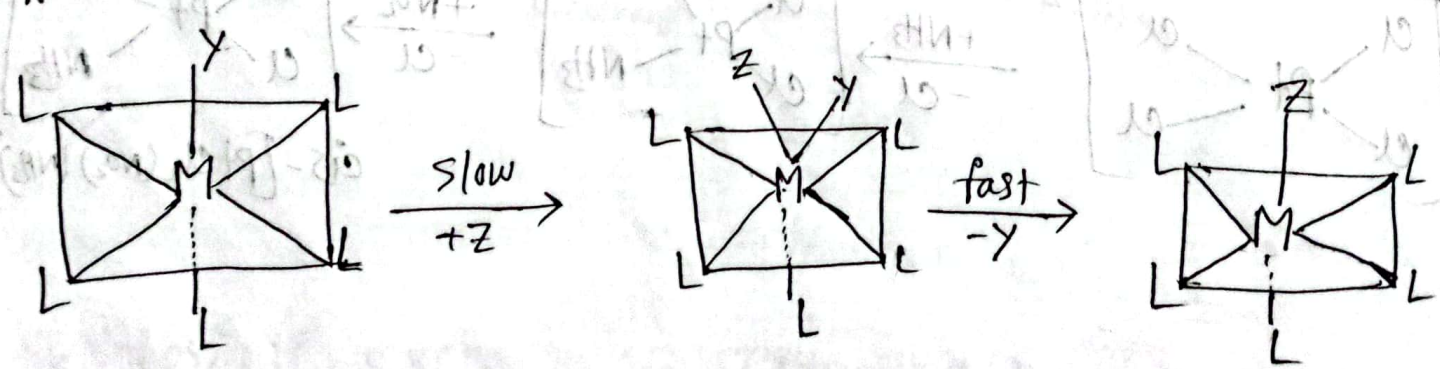
trans-$[PtCl_2(NH_3)(PPh_3)]$

2020(t)

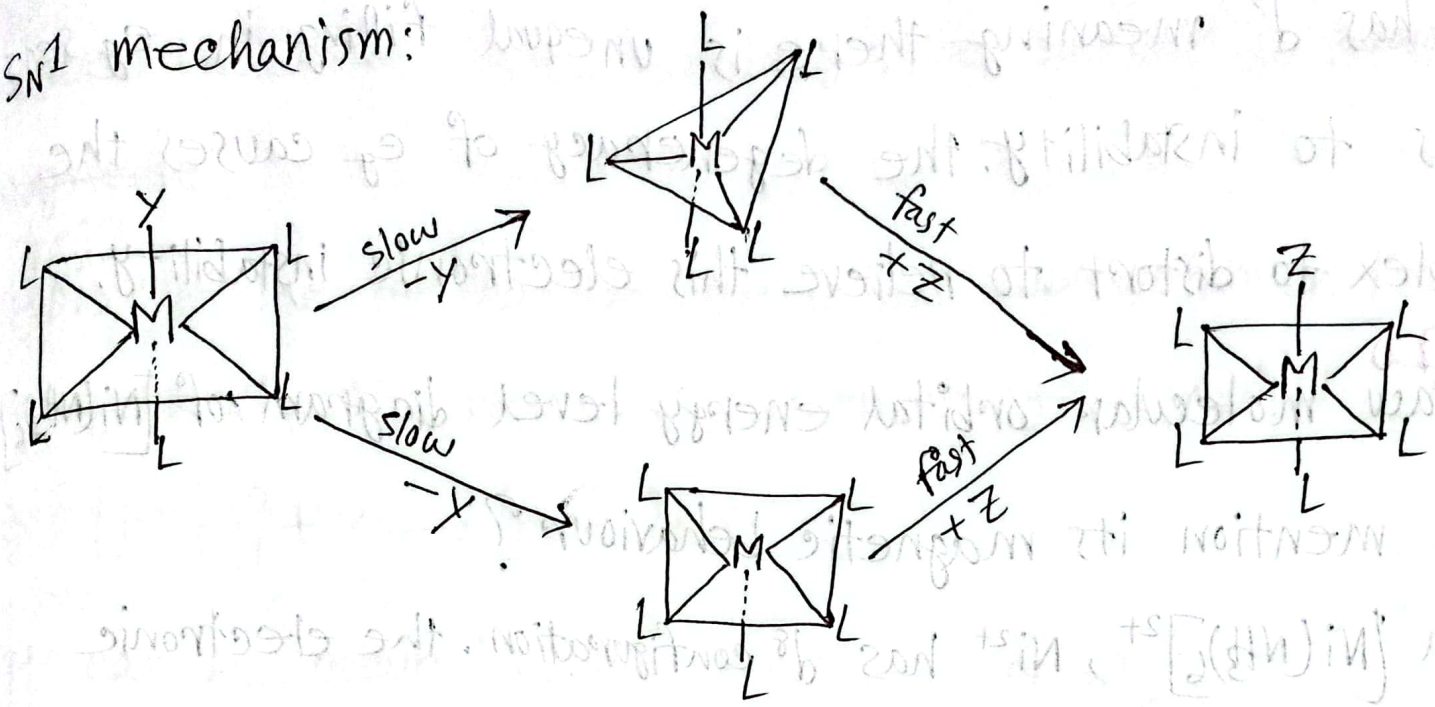
Discuss mechanism of S_N1 and S_N2 for octahedral complex

⇒ Let us ~~me~~ chemical reaction: $[L_5MY] + Z \longrightarrow [L_5MZ] + Y$

S_N2 mechanism:



S_N1 mechanism:



2020

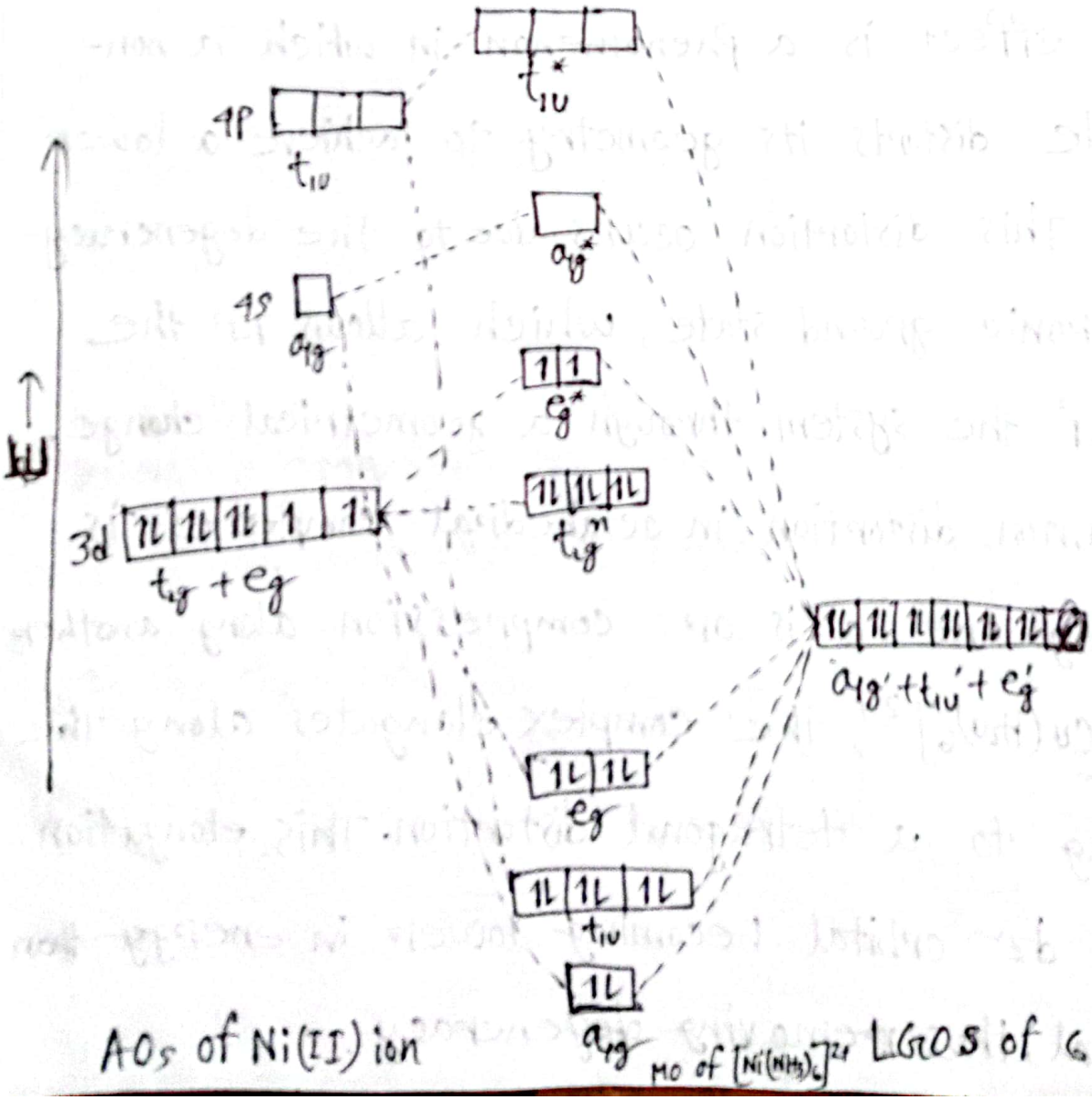
Explain Jahn-teller distortion in $[Cu(H_2O)_6]^{2+}$?

⇒ Jahn-teller effect is a phenomenon in which a non-linear molecule distorts its geometry to achieve a lower energy state. This distortion occurs due to the degeneracy of the electronic ground state, which allows for the stabilization of the system through a geometrical change. The most common distortion in octahedral complexes is elongation along one axis or compression along another. In case of $[Cu(H_2O)_6]^{2+}$, the complex elongates along the z-axis leading to a tetragonal distortion. This elongation results in the d_{z^2} orbital becoming lower in energy than the $d_{x^2-y^2}$ orbital, thus removing degeneracy.

Cu^{2+} has d^9 meaning there is unequal filling in e_g sets leads to instability. the degeneracy of e_g causes the complex to distort to relieve this electronic instability.

⊗ 2020 Draw molecular orbital energy level diagram of $[Ni(NH_3)_6]^{2+}$ and mention its magnetic behaviour?

⇒ In $[Ni(NH_3)_6]^{2+}$, Ni^{2+} has d^8 configuration. the electronic configuration $a_{1g}^2 t_{1u}^6 e_g^4 t_{2g}^6 e_g^2$



since two unpaired e⁻ remaining, the complex is paramagnetic.

~~Magnetic~~

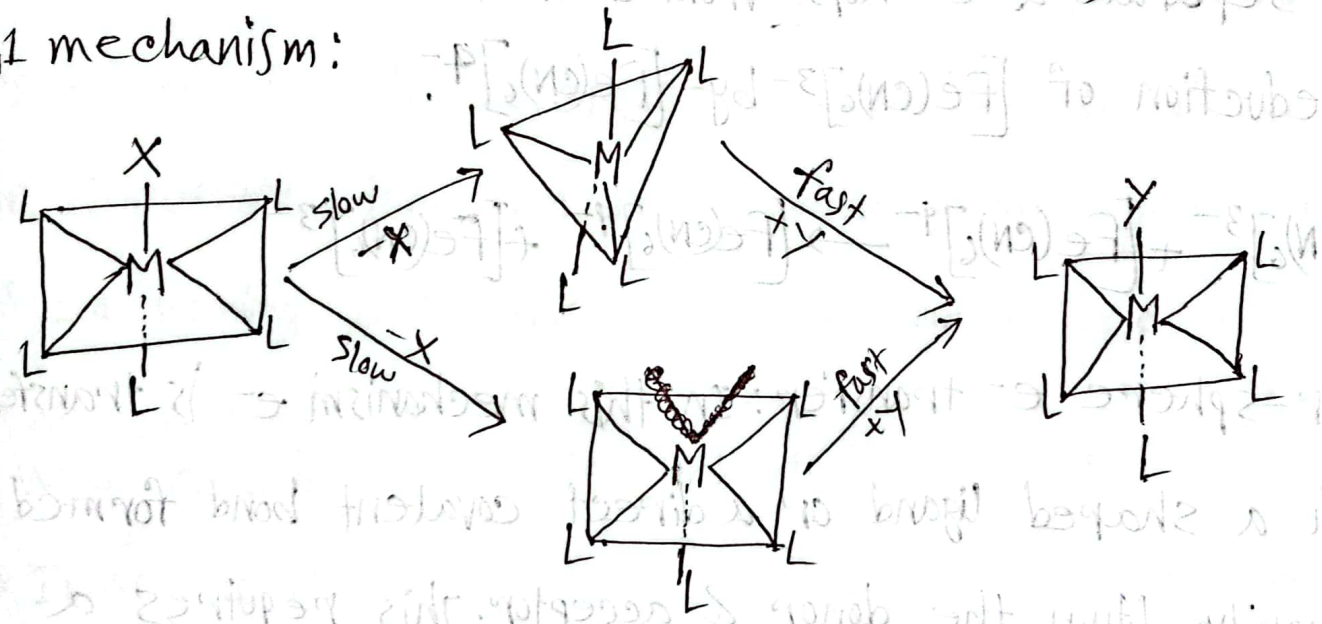
2019

⊛ Discuss the mechanism for ligand substitution in the complexes ML_5X, where y is entering ligand?

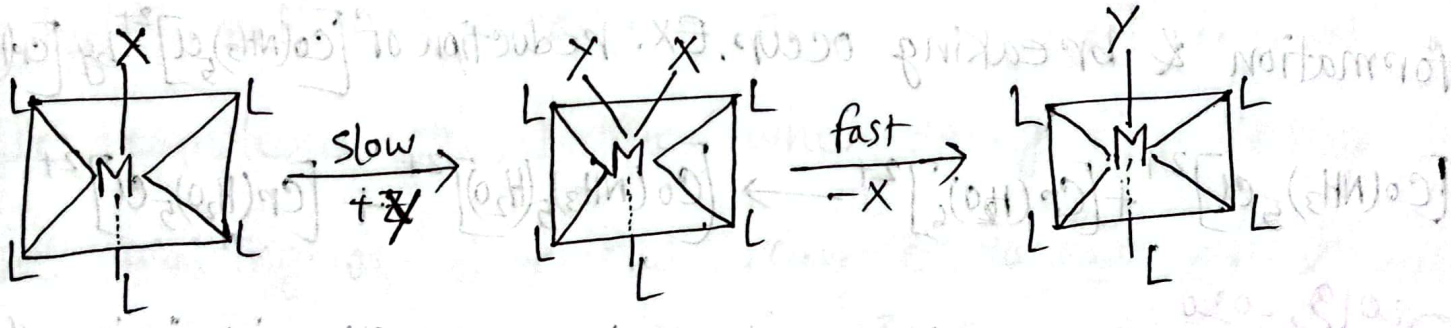
$$\Rightarrow [ML_5X] + Y \longrightarrow [ML_5Y] + X$$

It can be occur in two ways (S_N1 & S_N2).

S_N1 mechanism:



S_N2 mechanism:



⑤ 2019, 2020(H) Discuss two general mechanisms for e⁻ transfer reactions?

⇒ Two general mechanisms of e⁻ transfer reactions are outer-sphere & inner sphere e⁻ transfer.

① outer-sphere e⁻ transfer: This mechanism involves the transfer of e⁻ btwn two species with out significant structural change or formation of a chemical bond btwn them. The reactants remain separate & e⁻ 'hops' from donor to the acceptor.

Ex. reduction of $[Fe(CN)_6]^{3-}$ by $[Fe(CN)_6]^{4-}$.

$$[Fe(CN)_6]^{3-} + [Fe(CN)_6]^{4-} \longrightarrow [Fe(CN)_6]^{4-} + [Fe(CN)_6]^{3-}$$

② Inner-sphere e⁻ transfer: In this mechanism e⁻ is transferred through a shared ligand or a direct covalent bond formed temporarily btwn the donor & acceptor. this requires a chemical ~~reaction~~ interaction btwn the two reactants. Bond formation & breaking occur. Ex. reduction of $[Co(NH_3)_5Cl]^{2+}$ by $[Cr(H_2O)_6]^{2+}$

$$[Co(NH_3)_5Cl]^{2+} + [Cr(H_2O)_6]^{2+} \longrightarrow [Co(NH_3)_5(H_2O)]^{2+} + [Cr(H_2O)_5Cl]^{2+}$$

⑥ 2019, 2020 Anhydrous $AlCl_3$ is covalent but $AlCl_3 6H_2O$ is ionic-why?

⇒ In anhydrous form, $AlCl_3$ exist as a dimer with covalent bond btwn Al & Cl atoms. This dimeric structure is favored

due to the high polarizing power of the Al^{3+} ion. This high e^- density allows to ~~distort~~ polarizes the e^- cloud of the Cl^-, leading to a more covalent character.

when $AlCl_3$ dissolves in water, it undergoes hydration, & forms $[Al(H_2O)_6]^{3+}$.

$$AlCl_3 + 6H_2O \longrightarrow [Al(H_2O)_6]^{3+} + 3Cl^-$$

The water molecule stabilize the Al^{3+} ion by donating lone pairs of e^- to it & separates it from the Cl^- forming an ionic structure. Hydration reduces the polarizing power of the Al^{3+} so bonding btwn Al^{3+} and Cl^- becomes ionic in nature.

2018, 2015 (T)

(A) What are inner & outer orbital complexes?

⇒ Inner orbital complexes: involve the use of inner d-orbitals (n-1)d ~~in hybridi~~ of central metal atom/ion for hybridization. Typically involves d^2sp^3 hybridization. These complexes are formed when strong field ligands cause splitting of d-orbitals, forcing e^- to pair up in the lower-energy orbitals. Ex. $[Co(NH_3)_6]^{3+}$, $[Fe(CN)_6]^{4-}$ etc.

Outer orbital complexes: involve the use of outer d-orbital

nd of the central metal atom for hybridization. Typically involves sp^3d^2 hybridization. These complexes are formed when weak ligand cause only small splitting of d-orbitals, Ex. $[CoF_6]^{3-}$, $[Fe(H_2O)_6]^{3+}$.

2018
Ⓧ Give two example of inner & outer orbital complexes and explain the fact?

⇒ inner: Ex. $[Co(NH_3)_6]^{3+}$ & $[Fe(CN)_6]^{4+}$

① $[Co(NH_3)_6]^{3+}$: Co^{3+} undergo d^2sp^3 hybridization, involving inner 3d orbitals. The strong field ligand NH_3 cause for low spin & pairing of e^-, resulting complex is diamagnetic.

② $[Fe(CN)_6]^{4+}$: for strong ligand CN^-, we get octahedral, low spin, diamagnetic complexes.

outer: $[CoF_6]^{3-}$ & $[Fe(H_2O)_6]^{3+}$

① $[CoF_6]^{3-}$: Co^{3+} undergoes sp^3d^2, involving outer 4d orbitals. The weak ligand F^- cause for high spin & forms paramagnetic complexes.

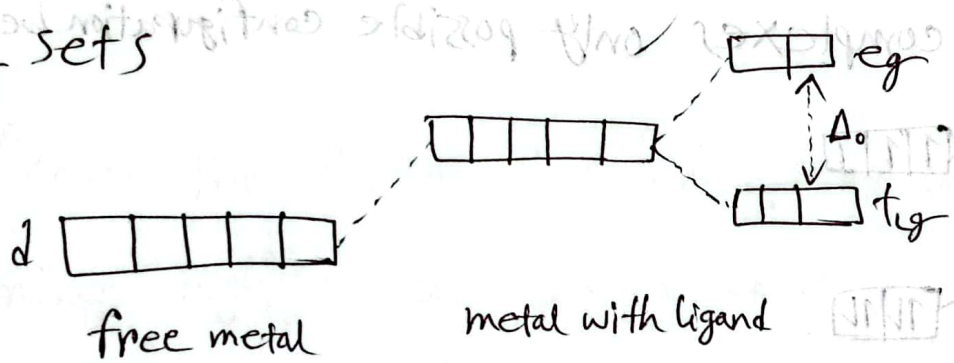
② $[Fe(H_2O)_6]^{3+}$: Fe^{3+} undergo sp^3d^2, involving outer 4d orbital. weak ligand cause for small splitting so ~~low sp~~ high

spin & paramagnetic complex.

(*) 2018, 2019(t) Outline how you apply CFT to explain five d-orbitals in an octahedral complexes are not degenerate?

⇒ CFT explains why the five d-orbitals of a central metal ion in an octahedral complexes are not degenerate.

In free metal ion five d-orbital are degenerate. The ligand interaction splits the d-orbits into two energy levels: e_g & t_{2g} sets



the orbitals $d_{x^2-y^2}$, d_{z^2} points directly along axes, experience strong repulsion from negatively charged ligand & energy increases significantly

(*) 2018, 2019(t) Explain Barycenter?

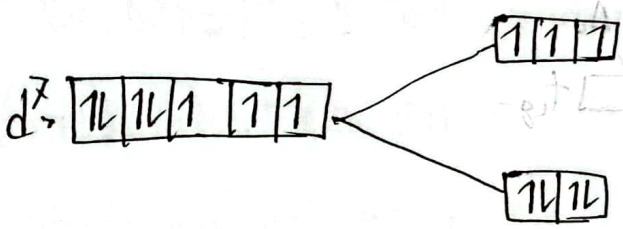
⇒ Barycenter is the center of mass of two or more bodies that orbit each other, such as a planet & its moon, or a star & its planets. It is the point where the masses of the system balance each other out, & it serves as the

center around which the bodies orbit. The barycenter is closer to the more massive body. position of barycenter can be calculated by the formula:

$$R = \frac{m_1 r_1 + m_2 r_2}{m_1 + m_2}$$

⊛ 2018, 2017 Calculate the spin only magnetic moment for d^7 ion in tetrahedral & ~~square planar~~ octahedral → 2017 complexes?

⇒ In tetrahedral complexes only possible configuration below:

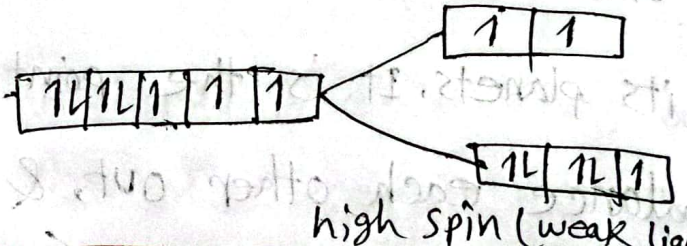


the energy difference btwn these sets is smaller compared to octahedral complexes, making tetrahedral generally high spin.

$$\therefore \mu_s = \sqrt{n(n+2)} \text{ BM} \quad [\text{where } n = \text{unpaired } e^-]$$

$$= \sqrt{3(3+2)} \text{ BM} = 3.87 \text{ BM}$$

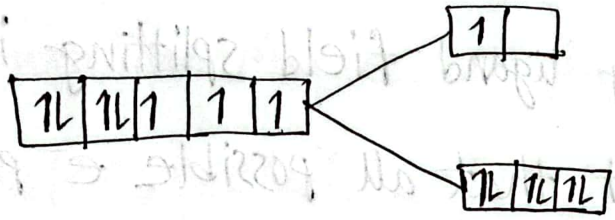
for octahedral complexes high & low spin possible.



high spin (weak ligan)

$\therefore \mu_s = \sqrt{3(3+2)}\ BM = 3.87\ BM$

for strong ligand low spin complex forms:



low spin (weak ligand)

$\mu_s = \sqrt{1(1+2)}\ BM = 1.73\ BM$

2018

(*) Discuss the factors that contribute to the preference for forming high or low spin d^4 complex?

(I) weak ligand favors High spin but In presence of Strong ligand Δ valve increase & favors low spin.

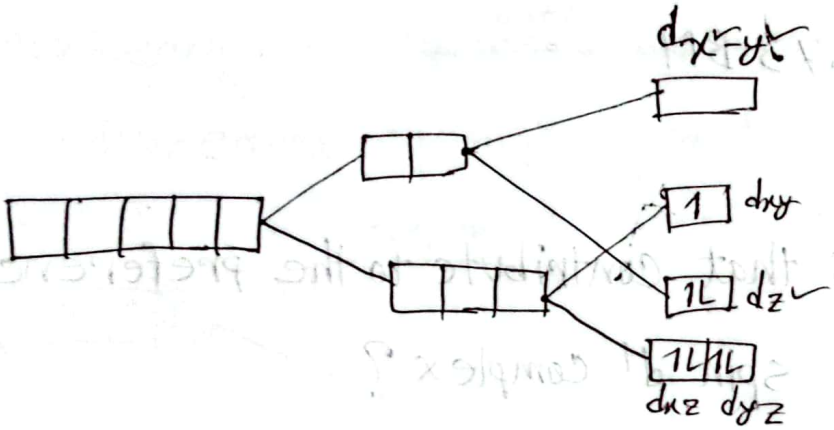
(II) Higher oxidation of metal ion cause for increasing Δ valve & prefer low spin.

(III) octahedral generally have larger Δ valve compared to tetrahedral, tend to low spin.

(IV) $P > \Delta$ it prefer high spin. $P < \Delta$ prefer low spin.

⊗ 2018 calculate spin magnetic moment(μ_s) for d^7 in square planar complexes?

⇒ A d^7 ion in a square planar is almost always a low-spin complex due to the strong ligand field splitting in square planar geometry. This meant that all possible e^- pairing occurs, leaving fewer or no unpaired e^-.



Here unpaired $e^- = 1$

$\therefore \mu_s = \sqrt{1(1+2)}\ BM = 1.73\ BM.$

⊗ 2017 what does the term ~~ligand~~ stereochemically inert lone pair? Give an example?

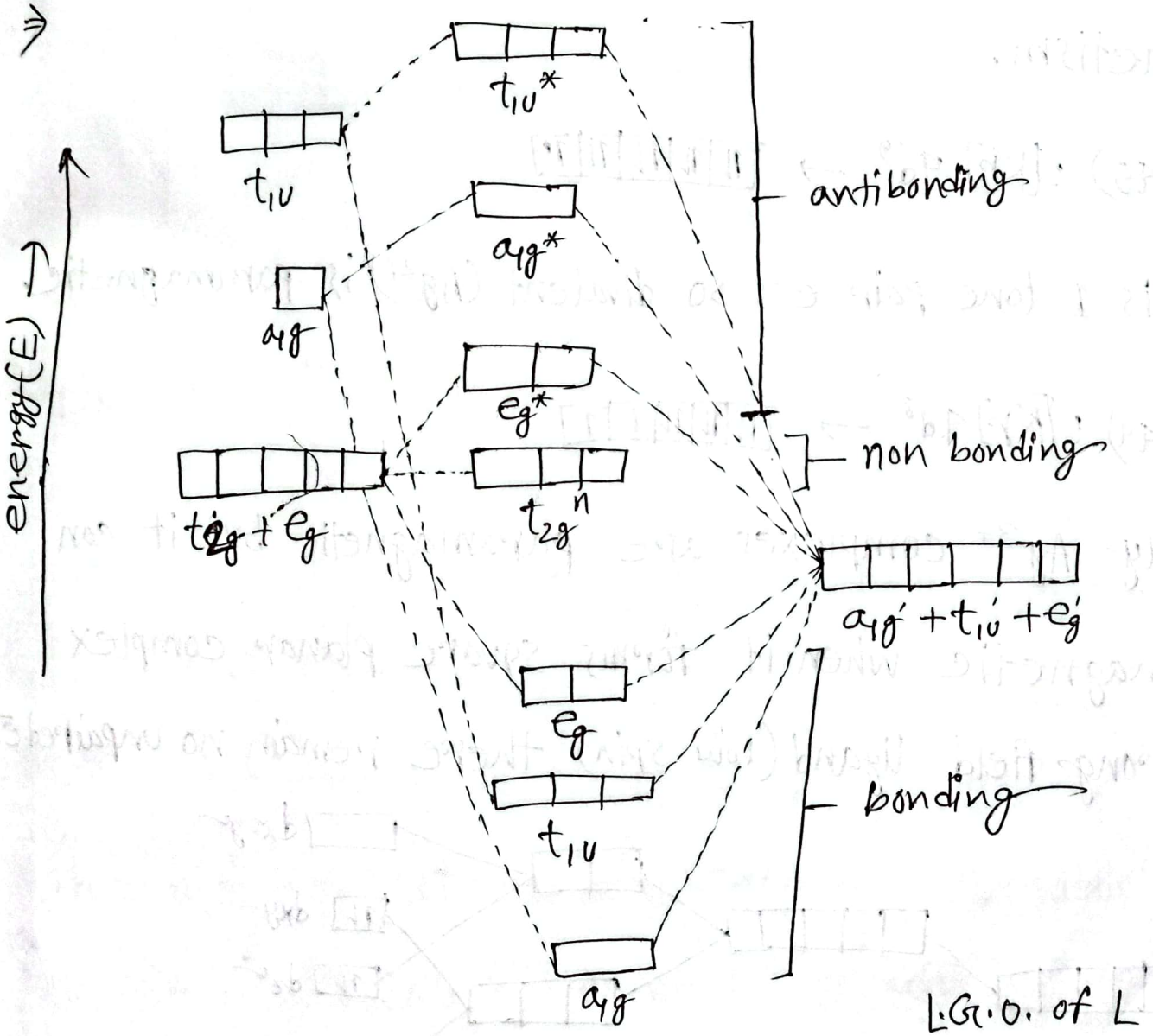
⇒ It is a lone pair of e^- that does not influence the molecular geometry of a compound. Unlike stereochemically active lone pairs, which repel bonding pairs & affect

the shape of a molecule. Inert lone pairs are typically localized in s-orbital, which are spherically-symmetric. This means they donot have a specific directional orientation, & thus they do not contribute to the molecule's overall shape.

EX. $[SbBr_6]^{3-}$, $[TeCl_6]^{2-}$, $[TeBr_6]^{2-}$, $[SbCl_6]^{2-}$.

(*) 2018, 2021, 2016
Draw M.O.D. for an octahedral complex $[CoL_6]^{3-}$?

⇒



(*) 2021 Complexes of Monovalent silver are ~~diamagg~~ diamagnetic, bivalent are para & trivalent Ag again diamagnetic. Explain?

⇒ $Ag(47) : [Kr]\ 4d^{10}\ 5s^1$

$Ag^+(46) : [Kr]\ 4d^{10} \longrightarrow$ ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿⇂

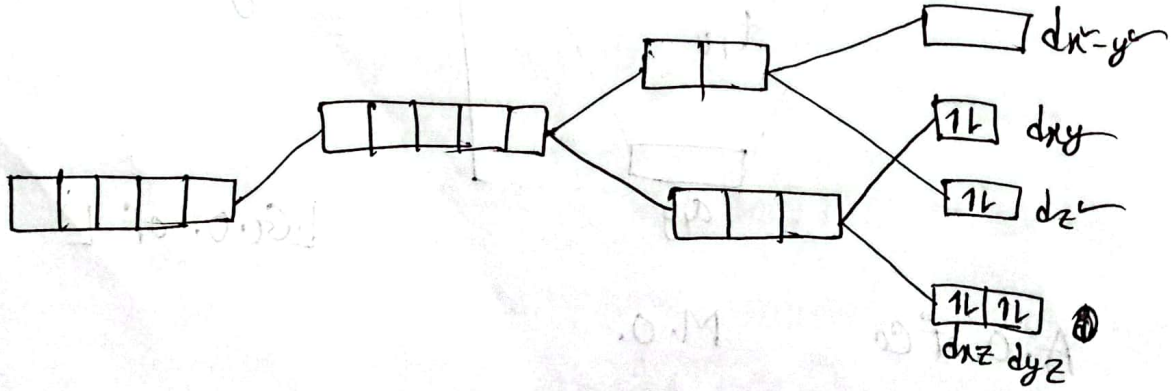
In monvalent, Ag^+ there is no lone pair so it shows diamagnetism.

$Ag^{2+}(45) : [Kr]\ 4d^9 \rightarrow$ ↿⇂ ↿⇂ ↿⇂ ↿⇂ ↿

there is 1 lone pair e^- so divalent (Ag^{2+}) is paramagnetic.

$Ag^{3+}(44) : [Kr]\ 4d^8 \rightarrow$ ↿⇂ ↿⇂ ↿⇂ 1 1

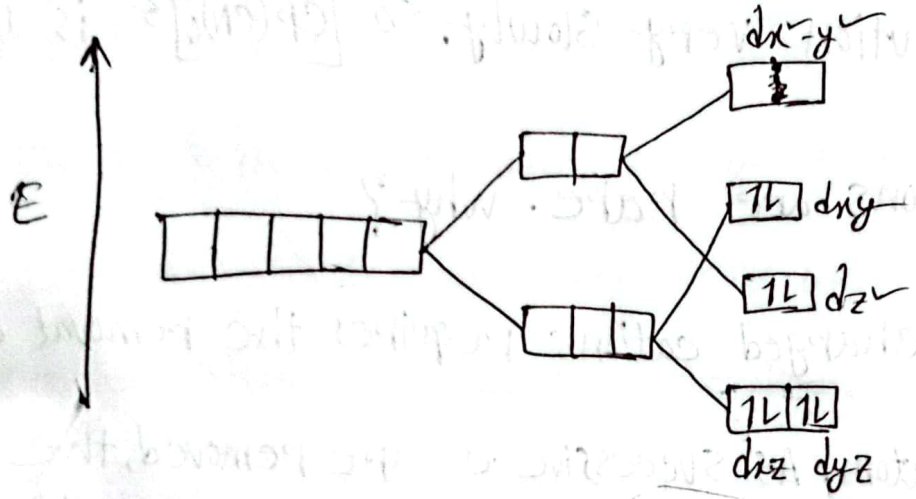
eventually Ag^{3+} complexes are paramagnetic but it can be diamagnetic when it forms square planar complex with strong field ligand (low spin) there remain no unpaired e



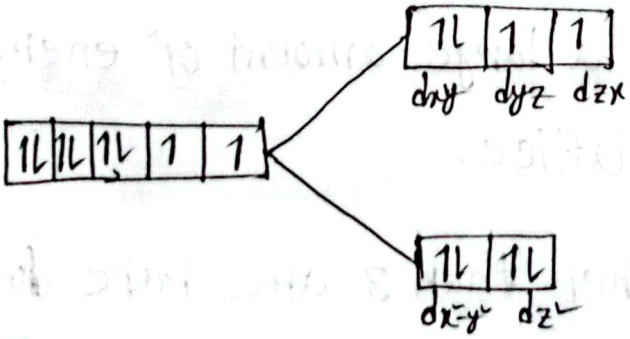
AS there is no unpaired e^- so it is diamagnetic.

2021
(#) The $[Ni(CN)_4]^{2-}$ ion, which has a square planar geometry, is diamagnetic, whereas the $[NiCl_4]^{2-}$ ion, which has a tetrahedral geometry, is paramagnetic. Show C.F. splitting diagram for those complexes?

⇒ for $[Ni(CN)_4]^{2-}$, Ni^{2+} contain d^8 configuration,



for $[NiCl_4]^{2-}$,



2021, 2017
(#) The high spin d^9 complex $[Cr(H_2O)_6]^{2+}$ is labile but the low spin d^9 $[Cr(CN)_4]^{2-}$ is inert - explain?

⇒ H_2O weak ligand so CFSE is relatively low. The low CFSE makes the complex more susceptible to ligand exchange

reactions, leading to its labile nature, meaning it can undergo substitution reactions quickly. so $[Cr(H_2O)_6]^{2+}$ is labile.

CN^- is strong ligand results high CFSE. The high CFSE makes the complex more stable & less likely to undergo ligand exchange reactions, leading to its inert nature, meaning it undergoes substitution very slowly. so $[Cr(CN)_6]^{3-}$ is inert.

2020
Highly charged cations are rare - why?

⇒ (I) Forming highly charged cations requires the removal of multiple e^- from an atom. As successive e^- are removed, the ionization energy increases significantly. This is because remaining e^- experience stronger electrostatic attraction from the nucleus.

(II) because they require a large amount of energy to form & create unstable ionic lattice.

(III) Ions with charges higher than 3 are rare due to high ionization energies & low resultant stability.

(IV) +ve charge increases, remaining e^- experience stronger electrostatic repulsion from each other. This repulsion destabilizes the ion & increases the energy required for further ionization.

CamScanner

2019, 2017

(*) The lability of a complex decreases with the increase of charge on central metal ion?

⇒ When charge on the central metal atom increases, its ability to attract & hold onto ligands increases due to the higher electrostatic interaction. This results in stronger metal-ligand bonds.

(ii) A central metal ion with a higher +ve charge also polarizes the ligands more effectively. This increases the strength of the interaction btwn the metal & the ligand, making complex more stable & less prone to dissociation (less labile).

(iii) with higher charge density, the central metal ion experiences greater ligand field stabilization, reducing the rate at which ligands can exchange with the surrounding environment. This decreases the lability of the complex.

2016, 2018

(*) How would you rationalize that the octahedral complex $[CrF_6]^{4-}$ has four Cr-F bonds of length 1.98 - 2.01 Å & two longer bonds of length 2.43 Å?

Cu-N, short, long, $[Cu(NH_3)_6]^{2+}$

⇒ It can be rationalized using Jahn-Teller distortion. Cr^{2+} has

d^4 configuration. d^4 fills the t_{2g} orbital & partially fills the e_g orbitals. The e_g orbitals are degenerate, meaning they have the same energy. The degeneracy of e_g makes the complex unstable. To remove degeneracy, the complex undergoes a geometric distortion. This distortion elongates two of the Cr-F bonds along the z-axis, while the other four bonds remain shorter. So Four Cr-F bonds have shorter lengths (1.98-2.01Å) due to ~~shorter~~ stronger bonding interactions. Two Cr-F bonds have longer length (2.13Å) due to weaker bonding interactions.

2016
⑧ $[AlF_6]^{3-}$, $[SiF_6]^{2-}$, $[PF_6]^-$ & SF_6 are all six coordinate complexes. Arrange them in order of decreasing ligand exchange rate & explain?

⇒ To arrange them, we need to consider:

① +ve charge increases on central atom increases the attraction btwn the central atom & the ligands, leading to stronger bonds & slower ligand exchange rates.

② Larger central atom generally have weaker Ligand bonds, leading to faster exchange rates.

(iii) strong ligand form more stable complexes & have slower exchange rates compared to weak field ligands.

Based on these factors, we can arrange the complexes as follows: $[AlF_6]^{3-} > [SiF_6]^{2-} > [PF_6]^- > SF_6$

because charge increase as follows $Al^{3+} < Si^{4+} < P^{5+} < S^{6+}$

and size Al > Si > P > S so ligand exchange order will be

$[AlF_6]^{3-} > [SiF_6]^{2-} > [PF_6]^- > SF_6$

2016

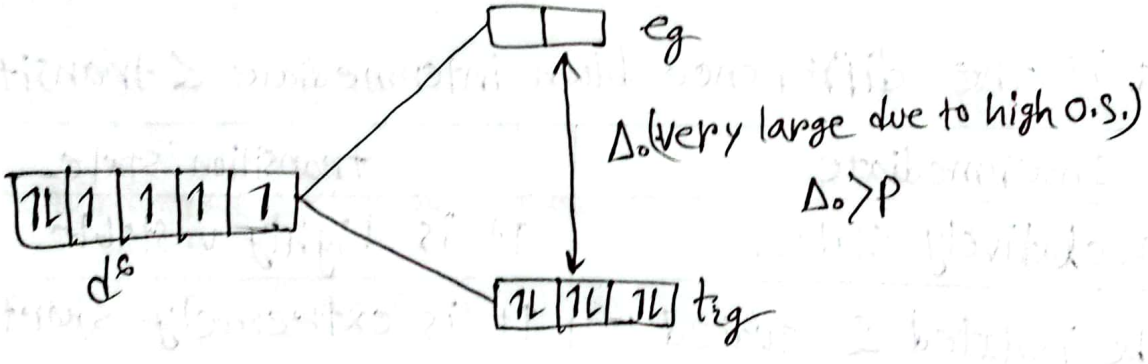
(x) what is the difference btwn intermediate & transition state?

Intermediate	transition state
It is Relatively stable	It is highly unstable
It can be isolated & studied	It is extremely short-lived.
It ~~ka~~ contain complete bonds	It contain partial bonds
It has definite geometry	It has indefinite geometry
It located minimum on the energy curve.	It located maximum on the energy curve
lower energy contain than T. state	contain higher energy than Intermediat
It acts as a 'stepping stone' btwn the reactant & products.	It represents the point at which bonds are breaking/forming simultaneously
It can be observed via spectroscopic or other analytical techniques.	It cannot be directly observed.

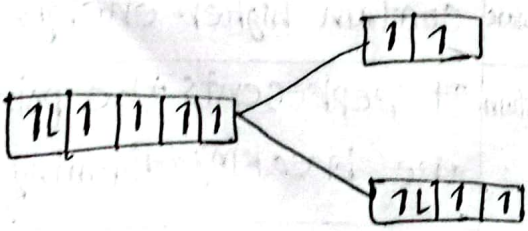
CamScanner

(★) 2021, 2017
$K_2[NiF_6]$ is diamagnetic while $K_3[CoF_6]$ is paramagnetic. Both have same d^6 configuration?

⇒ Due to the +4 oxidation state (High charge) of Ni in $K_2[NiF_6]$, it is low spin complex therefore no unpaired e^- & it is diamagnetic. due to high o.s. of Ni^{4+} attracts the F^- ligands with greater force of attraction, the ligand field splitting is significantly increased, Δ valve increased so low spin occured (Δ>P).



But in $K_3[CoF_6]$, Co is +3 oxidized & F^- is weak so Δo valve less than P.E. so high spin occured.



Unpaired e^- are present so it is Paramagnetic.

2021, 2019 (f)

(*) $[Ti(H_2O)_6]^{3+}$ exhibits a band with $\lambda_{max} = 510\,nm$. What color of light is absorbed & what color will aq. soln of $[Ti(H_2O)_6]^{3+}$ appear? Why?

⇒ The absorption spectrum of $[Ti(H_2O)_6]^{3+}$ exhibits a band with $\lambda_{max} = 510\,nm$. This means that the complex absorbs light at λ of 510 nm, which means soln absorbs green color. The soln will appear as the complementary color of the absorbed light. complementary color of green is purple or magenta. So Aq. soln of $[Ti(H_2O)_6]^{3+}$ will appear purple or magenta.

The color arises due to d-d electronic transitions in the $[Ti(H_2O)_6]^{3+}$ complex. Ti^{3+} in $[Ti(H_2O)_6]^{3+}$ has d^1 configuration. The six water ligands create an octahedral crystal field around Ti^{3+}, splitting the d-orbitals into t_{2g} & e_g sets. The absorption of light at 510 nm corresponds to the excitation of the d-electron from the lower energy t_{2g} level to the higher energy e_g level. That's why complementary color of green will apper which is purple or magenta.

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2021

Explain the effect on the d-orbital energies when an octahedral complex is compressed along the z-axis?

⇒ In a regular octahedral field, d-orbital split into ~~two~~ t_{2g} and e_g sets.

But when the complex compressed along the z-axis, the ligands along the z-axis are closer to the metal center, increasing the repulsion along that axis. conversely, the ligands along the x- and y-axes are farther away, decreasing the repulsion in those directions. d_{z^2} Experiences the greatest increase in energy due to stronger repulsion from the ligands closer along the z-axis. $d_{x^2-y^2}$ experiences a decrease in energy because the ligands along x & y-axes are farther away. energies of d_{xz}, d_{yz} increase slightly due to z-axis compression while d_{xy} remains relatively unaffected because it lies in xy plane.

2018

What two factors contribute to giving high trans effects?

⇒ Two primary factors contribute to a high trans

effect: ① Strong σ-donor ability: Ligands that are strong σ-donor donate e⁻ density effectively to the metal center through a σ-bond. This increased e⁻ density at the metal center weakens the bond trans to the strong σ-donor, making it more susceptible to substitution. Ex. CH_3^-, H^-

② Strong π-acceptor ability: Ligands that are strong π-acceptors can withdraw e⁻ density from the metal center through π-backbonding. This reduces the e⁻ density at the metal center, weakining the bond trans to the π-accept ligand. Ex. CO, CN^-, NO_2^-.

2017

✱ of the first-row transition metal complexes $[M(NH_3)_6]^{3+}$. which metals are predicted by the Jahn teller theorem to have distorted complexes?

⇒ For $[M(H_2O)_6]^{3+}$, the relevant metals in the first row transition series are M = Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu & Zn among them Mn^{3+} & Cu^{2+} are predicted by the Jahn-teller theorem to have distorted complexes of the formula $[M(NH_3)_6]^{3+}$.

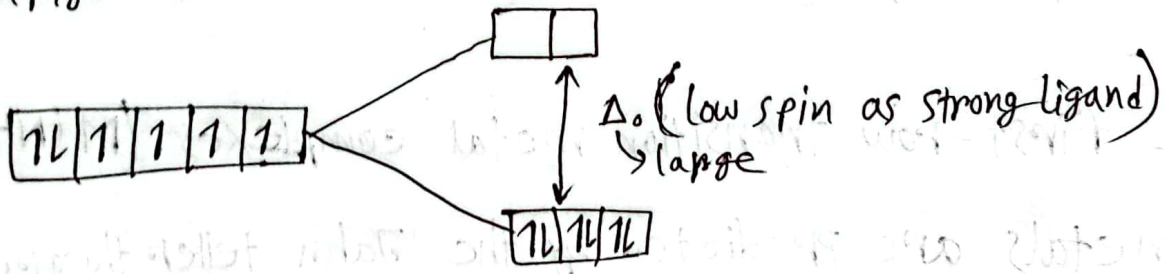
① Mn^{3+} has d^4 configuration, Its high spin configuration has two

unpaired e^- in the e_g orbitals. The complex will undergo a tetragonal distortion to remove this degeneracy, leading to a lower energy state.

Cu^{2+} has d^9 configuration; It has 3 e^- in e_g degenerate orbitals. The complex will also undergo a tetragonal distortion to remove degeneracy, leading to a lower-energy state.

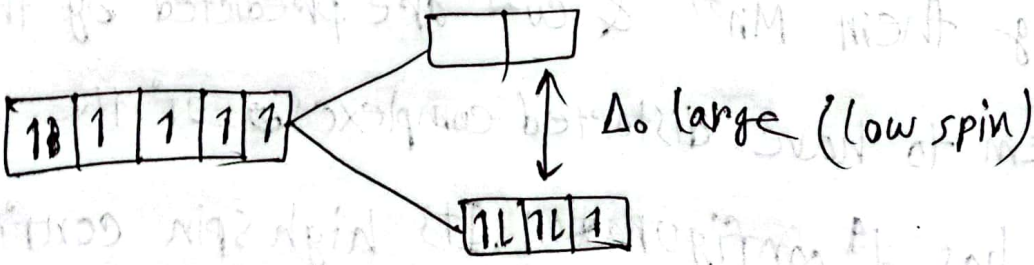
⊗ 2015(t) Give the number of unpaired e^- & their magnetic behaviour: $[Fe(CN)_6]^{4-}$ & $[Fe(CN)_6]^{3-}$. ?

⇒ In $[Fe(CN)_6]^{4-}$, oxidation number of Fe is +2. So it has d^6 configuration.



∴ unpaired e^- = 0, magnetism = diamagnetic

In $[Fe(CN)_6]^{3-}$, oxidation number of Fe is +3. So it has d^5 configuration.



unpaired e^- = 1, magnetism: paramagnetic

2015 (H)

(*) write important features of CFT?

⇒ (I) The central metal cation is surrounded by ligands which contain one or more lone pair of electrons.

(II) the ionic ligands (e.g. F^-, Cl^-, CN^- etc) are regarded as -ve point charges (also called point charges) and the neutral ligands (eg. H_2O, NH_3 etc) are regarded as point dipoles.

(III) If the ligand is neutral, the -ve end of this ligand dipole is oriented towards metal cation.

(IV) The CFT does not provide for e^- to enter the metal orbitals. Thus the metal ion & the ligands do not mix their orbitals or share e^-, i.e. it does not consider any orbital overlap.

(V) According to CFT, the bonding btwn metal cation & ligand is not covalent but it is regarded as purely electrostatic attraction btwn +ve & -ve charged species.

Extra

(*) write salient feature of CFT?

⇒ (I) Focuses on the d-orbitals of the metal.

(II) The central metal cation is surrounded by ligands which contain one or more lone pairs of e^-.

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(iii) The ionic ligands (F^-, Cl^-, CN^-) are regarded as negative point charge & the neutral ligands (H_2O, NH_3) regarded as point dipoles.

(iv) If the ligand is neutral, the -ve end of this ligand dipole is oriented towards metal cation.

(v) The interaction btwn the metal ~~ion~~ cation & the ligands is regarded as purely electrostatic i.e. the metal-ligand bond is considered to be 100% ionic.

(vi) Electrostatic interactions in a complex btwn +ve metal ion & -ve charges of ligands treats ligands as point charges.
If the ligand is -ve charged → Ion-ion interaction
If the ligand is neutral → Ion-dipole interaction.

(vii) provides stability & holds complex together.

(viii) Repulsion btwn the lone pair of e^- on the ligand & the e^- in the d-orbital of the metal ion.

(ix) This influences the d-orbital energies.

(x) 2020(t)
which ion is more stable: Cu^{2+} or Cu^+?

⇒ Cu^{2+} is more stable than Cu^+ in aqueous sln due to

its higher hydration energy. This means more energy is released when Cu^{2+} bonds with water molecule, making it more stable. Cu^{2+} has higher charge density than Cu^{+}, which allows it to form stronger bonds with H_2O & release more energy. Cu^{+} is unstable in aqueous soln due to disproportion reaction

$$2Cu^{+} \longrightarrow Cu^{2+} + Cu$$

2020(t)

Which ion is more difficult to oxidize - why? Mn^{2+} or Fe^{2+}?

⇒ Mn^{2+} is more difficult to oxidize than Fe^{2+}. because Mn^{2+} has d^5 means half filled d-orbital. Half-filled & full filled d-orbitals are more stable due to exchange energy. But Fe^{2+} has d^6, which is less stable.

2020(t)

Why transition elements have high hydration energy?

⇒ (i) Transition element have higher charge. This higher charge creates a strong electrostatic attraction btwn the ion & water molecules, increasing the hydration energy.

(ii) Transition metals are relatively smaller compared to other element of the same period. Smaller ion have higher charge density which leads to stronger electrostatic

attraction btwn ion & water molecules.

(III) Transition elements have a high effective nuclear charge which means the nucleus exerts a strong pull on the e⁻. This strong pull leads to a smaller ionic size & higher charge density, further contributing to high hydration energy.

(IV) Transition metal ions have a high polarizing power, which means they can distort the e⁻ cloud of water molecules. this distortion leads to higher hydration energy.

(V) H_2O can act as ligands, donating e⁻ density to the empty or partially filled d-orbitals of the metal ion & stabilize the hydrated complex.

2020(I)

(8) Define: (I) Hydration energy (II) Electron affinity (III) Ionization energy (IV) Pseudo Transition metals ?

(I) Hydration energy: The amount of heat released when one mole of gaseous ions dissolve in water to form an infinite dilute solution. This is always an exothermic process. Denoted by ΔH_h.

EX: $H^+ + H_2O \longrightarrow H_3O^+$; $\Delta H_h = -1075\ KJmol^{-1}$.

(II) Electron affinity: The amount of energy that is release

when an e^- is added ~~to an atom~~ in the outermost shell of an isolated gaseous atom.

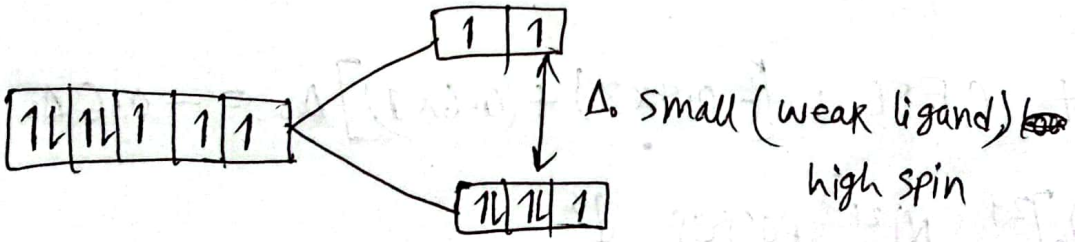
$M(g) + e^- \rightarrow M^+(g)$

(III) Ionization energy: The amount of energy required to remove the highest-energy e^- from an isolated atom in the gaseous state. $A(g) \longrightarrow A^+(g) + e^-$

(IV) pseudo transition metal: are elements that are often discussed with transition metals because they have similar properties, but are not technically transition metals. EX. Zn, Cd, ~~Hg~~

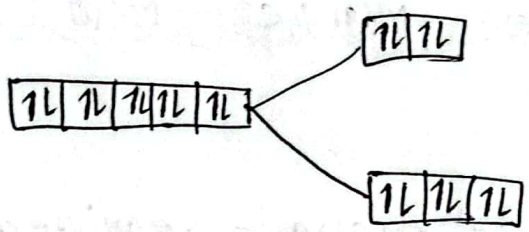
(※) 2022, 2024(t) Determine unpaired e^- & CFSE value of: ① $[CoCl_4]^{2-}$ ② $[Zn(CN)_6]^{4-}$ ③ $[Fe(H_2O)_6]^{3+}$ ④ $[Cr(H_2O)_6]Cl_2$ ⑤ $[Ni(NH_3)_6]^{2+}$ ⑥ $K_3[Co(CN)_6]$?

⇒ ① for $[CoCl_4]^{2-}$, Co's oxidation number = +2, it contain d^7



∴ upaired = 3, CFSE = $[(-0.4\times5)+(0.6\times2)]\,\Delta_o = -0.8\,\Delta_o$

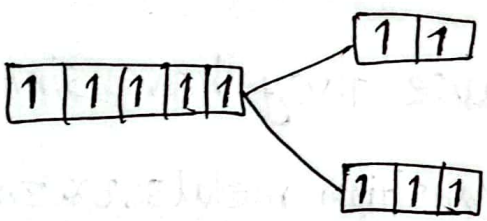
(II) for $[Zn(CN)_6]^{4-}$, Zn^{2+} posses d^{10}.



(strong ligand low spin)

unpaired = 0 , CFSE = $[(-0.4\times6)+(0.6\times4)]\Delta_o = 0$

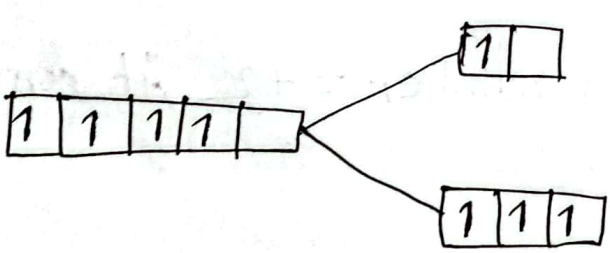
(III) for $[Fe(H_2O)_6]^{3+}$. Fe^{3+} posses d^5.



(weak ligand high spin)

unpaired = 0 , CFSE = $[(-0.4\times3)+(0.6\times2)]\Delta_o = 0$

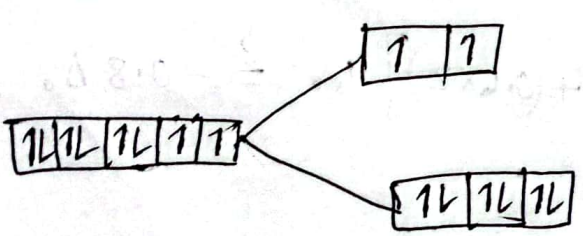
(IV) for $[Cr(H_2O)_6]Cl_2$, Cr^{2+} posses d^4



(weak ligand high spin)

unpaired = 4, CFSE = $[(-0.4\times3)+(0.6\times1)]\Delta_o = -0.6\Delta_o$

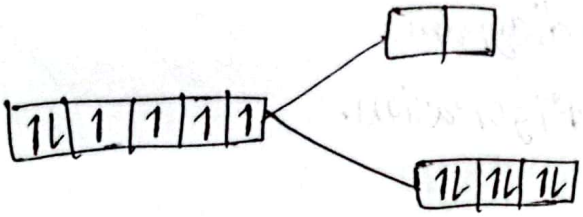
(V) for $[Ni(NH_3)_6]^{2+}$, Ni^{2+} posses d^8



(strong ligand low spin)

unpaired = 2 , CFSE = $[(-0.4\times6)+(0.6\times2)]\Delta_o$ ~~$= -1.8\Delta_o$~~ $= -1.2\Delta_o$

(VI) for $K_3[Co(CN)_6]$, Co^{3+} posses d^6



(Strong ligand low spin)

unpaired = 0 , $CFSE = [(-0.4 \times 6) + (0.6 \times 0)]\Delta_o = -2.4\Delta_o$

(X) 2022 calculate spin only magnetic moment for Mn^{3+} & Fe^{3+}?

$\Rightarrow Mn^{3+} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^4$

unpaired e^-, $n = 4$

$\therefore \mu_s = \sqrt{4(4+2)} = 4.899$ BM.

for Fe^{3+},

$Fe^{3+} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

unpaired e^- $n = 5$

$\therefore \mu_s = \sqrt{5(5+2)} = 5.92$ BM.